CURRICULUM

(REGULATIONS 2023)

FOR

B.E. COMPUTER SCIENCE AND ENGINEERING

CHOICE BASED CREDIT SYSTEM

(Applicable to the students admitted from the Academic Year 2023-24 onwards)



EASWARI ENGINEERING COLLEGE
(Autonomous Institution)
Bharathi Salai, Ramapuram, Chennai - 600 089

[A Unit of SRM Group of Educational Institutions, Approved by AICTE | Affiliated to Anna University, Chennai | NAAC Accredited 'A' Grade | 2(f) & 12(B) Status (UGC) | ISO 9001:2015 Certified | NBA Accredited Programmes | FIST Funded (DST) | SIRO Certified (DSIR)]

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SEMESTER I									
S. No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231LEH101T	Technical English	HS	3	-	-	-	3	40 / 60
2	231MAB101T	Matrices and Calculus	BS	3	1	-	-	4	40 / 60
3	231PYB101T	Engineering Physics	BS	3	-	-	-	3	40 / 60
4	231CYB101T	Engineering Chemistry	BS	3	-	-	-	3	40 / 60
5	231GES104T	Problem Solving through Python Programming	ES	2	-	-	-	2	40 / 60
6	231GEH101T	Heritage of Tamils / தமிழர்மரபு	HS	1	-	-	-	1	40 / 60
LABORATORY									
8	231CYB011L	Chemistry Laboratory	BS	-	-	2	-	1	60 / 40
9	231GES011L	Basic Workshop Practice	ES	-	-	2	-	1	60 / 40
10	231GES013L	Basic Electrical and Electronics Engineering Laboratory	ES	-	-	4	-	2	60 / 40
11	231GES114L	Python Programming Laboratory	ES	-	-	3	1	2	60 / 40
12	231GEH111L	Induction Training&	MC	-	-	2	-	0	-
TOTAL CREDITS								**22**	**-**

& - Mandatory to complete the course

SEMESTER II									
S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231LEH201T	Professional Communication	HS	3	-	-	-	3	40 / 60
2	231MAB201T	Advanced Calculus and Complex Analysis	BS	3	1	-	-	4	40 / 60
3	231PYS202T	Physics for Information Science	ES	3	-	-	-	3	40 / 60
4	231GEB201T	Environmental Science for Computing Sciences	BS	2	-	-	-	2	40 / 60
5	231CSC201T	Programming in C	PC	3	-	-	-	3	40 / 60
6	231GEH201T	தமிழரும் ததொழில் நுட்பமும் /Tamils and Technology	HS	1	-	-	-	1	40 / 60
LABORATORY									
7	231PYB011L	Physics Laboratory	BS	-	-	2	-	1	60 / 40
8	231GES012L	Visualization of Design and Drawing	ES	-	-	3	1	2	60 / 40
9	231CSC211L	C Programming Laboratory	PC	-	-	3	1	2	60 / 40
TOTAL CREDITS								**21**	**-**

SEMESTER III

S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231MAB302T	Discrete Mathematics	BS	3	2	-	-	4	40 / 60
2	231CSS301T	Digital Principles and Computer Organization	ES	3	-	-	-	3	40 / 60
3	231CSC301T	Data Structures and Algorithms	PC	3	-	-	-	3	40 / 60
3	231CSC302T	Object Oriented Programming with C++	PC	3	-	-	-	3	40 / 60
5	231CSC303T	Foundations of Data Science	PC	3	-	-	-	3	40 / 60
LABORATORY									
6	231CSC311L	Data Structures Laboratory	PC	-	-	4	0	2	60 / 40
7	231CSC312L	Object Oriented Programming Laboratory	PC	-	-	3	1	2	60 / 40
8	231CSC313L	Data Science Laboratory	PC	-	-	4	0	2	60 / 40
9	231SDA001	Comprehensive Programming- Foundations	EEC	0	0	2	0	0	Completed / Not completed
TOTAL CREDITS								**22**	**-**

^ Mandatory to register for the course and earn one credit
& Mandatory to attend Induction training programme and earn one credit.

SEMESTER IV

S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231MAB403T	Probability And Number Theory	BS	3	1	-	-	4	40 / 60
2	231CSC401T	Database Management Systems	PC	3	-	-	-	3	40 / 60
3	231CSC402T	Design and Analysis of Algorithms	PC	3	-	-	-	3	40 / 60
4	231CSC403T	Operating Systems	PC	3	-	-	-	3	40 / 60
5	231CSC404T	Java Programming	PC	3	-	-	-	3	40 / 60
LABORATORY									
6	231CSC411L	Database Management Systems Laboratory	PC	-	-	4	0	2	60 / 40
7	231CSC412L	Operating Systems Laboratory	PC	-	-	4	0	2	60 / 40
8	231CSC413L	Java Programming Laboratory	PC	-	-	3	1	2	60 / 40
		Skill Development / Value Added Course (Branch specific)	MC	0	0	2	0	0	Completed / Not completed

								TOTAL CREDITS	22	-

SEMESTER V									
S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231CSC501T	Computer Networks	PC	3	-	-	-	3	40/60
2	231CSC502T	Software Engineering	PC	3	-	-	-	3	40/60
3	231CSC503J	Artificial Intelligence and Machine Learning	PC	3	-	2	-	4	40/60
4		Professional Elective – I	PE	3	-	-	-	3	40/60
5		Professional Elective – II	PE	3	-	-	-	3	40/60
6		Professional Elective – III	PE	3	-	-	-	3	40/60
LABORATORY									
7	231CSC511L	Computer Networks Laboratory	PC	-	-	3	1	2	60 / 40
8	231CSC512L	Innovation Lab / Mini project / Design project / Multidisciplinary project	EEC	-	-	1	1	1	100/0
9		Skill Development / Value Added Course (Branch specific)	MC	-	-	2	0	0	Completed / Not completed
10		Mandatory Course - I	MC	-	-	-	-	0	Completed / Not completed
							TOTAL CREDITS	22	-

SEMESTER VI									
S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231CSC601T	Web Technologies	PC	3	0	0	0	3	40 / 60
2	231CSC602T	Principles of Compiler Design	PC	3	1	0	0	4	40 / 60
3		Professional Elective – IV	PE	3	0	0	0	3	40/60
4		Professional Elective – V	PE	3	0	0	0	3	40/60
5		Professional Elective - VI	PE	3	0	0	0	3	40/60
6		Open Elective – I – (Basic Science)	OE	3	0	0	0	3	40/60
7	231CSA601T	Comprehensive Examination*	MC	0	0	0	0	3*	100/0
8		Mandatory Course - II	MC	2	0	0	0	0	Completed / Not completed
LABORATORY									
9	231CSC611L	Web Technologies Laboratory	PC	-	-	4	0	2	60 / 40
10	231CSC612L	Mobile Application Development Laboratory	PC	-	-	3	1	2	60 / 40

		Skill Development / Value Added Course (Branch specific)	MC	0	0	2	0	0	Completed / Not completed
								TOTAL CREDITS	**23**	**-**

SEMESTER VII

S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231CSC701T	Cryptography and Information Security	PC	3	-	-	-	3	40 / 60
2	231GEH701T	Human Values and Ethics	HS	3	-	-	-	3	40/60
3		Management Elective	HS	3	-	-	-	3	40/60
4		Professional Elective - VII	PE	3	-	-	-	3	40/60
5		Open Elective - II	OE	3	-	-	-	3	40/60
6		Open Elective – III	OE	3	-	-	-	3	40/60
LABORATORY									
7	231CSC711L	Cryptography and Information Security Laboratory	PC	-	-	3	1	2	60 / 40
8		Project Phase I	EEC	0	0	0	1	1	60/40
9		Industrial Training / Internship	EEC	0	0	0	0	2	100/0
							TOTAL CREDITS	**23**	**-**

SEMESTER VIII

S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
LABORATORY									
1	231CSP811J	Project Work / Internship based Project	EEC	0	0	0	8	8	60/40
							TOTAL CREDITS	**8**	**-**

Total: **(163)**

Mandatory Course – I

Course Code	Course Title	Hours / Week				Credit	Internal / External %
		L	T	P	R		
231GEM501T	NSS	-	-	3	-	0	Completed / Not Completed
231GEM502T	YRC	-	-	3	-	0	
231GEM503T	Yoga Practice	-	-	3	-	0	
231GEM504T	Sports Activities	-	-	3	-	0	

Mandatory Course – II

Course Code	Course Title	Hours / Week				Credit	Internal / External %
		L	T	P	R		
231GEM601T	Constitution of India	3	-	-	-	0	100 / 0
231GEM602T	State, Nation Building and Politics in India	3	-	-	-	0	100 / 0
231GEM603T	Disaster Risk Reduction and Management	3	-	-	-	0	100 / 0
231GEM604T	Industrial Safety	3	-	-	-	0	100 / 0
231GEM605T	Gender Studies	3	-	-	-	0	100 / 0
231GEM606T	History of Science and Technology In India	3	-	-	-	0	100 / 0

Management Elective

Course Code	Course Title	Hours / Week				Credit	Internal / External %
		L	T	P	R		
231MBH701T	Principles of Management	3	-	-	-	3	40 / 60
231MBH702T	Total Quality Management	3	-	-	-	3	40 / 60
231MBH703T	Engineering Economics and Financial Accounting	3	-	-	-	3	40 / 60
231MBH704T	Human Resource Management	3	-	-	-	3	40 / 60
231MBH705T	Knowledge Management	3	-	-	-	3	40 / 60
231MBH706T	Industrial Management	3	-	-	-	3	40 / 60

CREDIT DISTRIBUTION

SEMESTER	I	II	III	IV	V	VI	VII	VIII	CREDIT
Humanities and Social Sciences (HS)	4	4					6		**14**
Basic Sciences(BS)	11	7	4	4					**26**
Engineering Sciences (ES)	7	5	3						**15**
Professional Core (PC)		5	15	18	12	11	5		**66**
Professional Electives (PE)					9	9	3		**21**
Open Electives (OE)						3	6		**09**
Employability Enhancement Courses (EEC)					1		3	8	**12**
Total Credit	**22**	**21**	**22**	**22**	**22**	**23**	**23**	**8**	**163**

SEMESTER I

231LEH101T	TECHNICAL ENGLISH (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		3	0	0	0	3

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To improve the communicative competence of learners
2.	To help learners use language effectively in academic /work contexts
3.	To build on students' English language skills by engaging them in listening, speaking and grammar learning activities that are relevant to authentic contexts
4.	To develop learners' ability to read and write complex texts, summaries, articles, blogs, definitions, essays and user manuals.
5.	To use language efficiently in expressing their opinions via various media.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Listen and comprehend lectures and talks in their area of specialization successfully.	K2
CO2:	Speak appropriately and effectively in varied formal and informal contexts.	K4
CO3:	Read technical texts.	K5
CO4:	Write Letters , Paragraphs, Descriptions.	K6
CO5:	Speak convincingly and express opinions on technical topics.	K6
CO6:	Communicate effectively through emails and analyze issues , technical articles and involve in speed writing.	K4

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	-	-	-	-	-	-	-	-	1	1	-	-
CO2	-	-	-	-	-	-	-	-	1	2	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	1	-	1
CO6	-	-	-	-	-	-	-	-	1	-	-	-
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

UNIT	**TITLE**	**PERIODS**
I		**9**

Listening : Active listening -Types – Barriers **Speaking**: Self – introduction and introducing others / short conversation in formal and informal contexts- **Reading** : skimming and scanning – Intensive and extensive reading -**Writing**-Developing an outline– Developing hints **Vocabulary development and Grammar**: Parts of speech – Word formation – One word substitution – Phrasal verbs.

UNIT	**TITLE**	**PERIODS**
II		**9**

Listening: Listening to Telephonic conversation – Etiquettes- Do's and Don'ts - **Speaking**:– Making technical and presentation- Presentation Do's and Don'ts - **Reading**:- Interpretative and critical levels of reading – **Writing** – Technical writing – letter (request/ permission/ complaint)- **Vocabulary and Grammar** –Tenses – Subject verb agreement –Pronoun (Possessive and Relative Pronoun)

UNIT	**TITLE**	**PERIODS**
III		**9**

Listening: Listening to longer technical talks - **Speaking**: Expressing opinions – one to one debate – **Reading**: Comprehension- Pre- reading- post- reading - **Writing**- Email writing – Etiquette- Instructions and Recommendations - **Vocabulary and Grammar**: Adverbs – prepositions - Degrees of comparison (Comparative Degree)

UNIT	**TITLE**	**PERIODS**
IV		**9**

Listening: Listening to speeches of formal and informal conversation and answer comprehension -**Speaking**: Sharing information of a personal kind – greeting- taking leave – routine action - Reading: Note making and summarizing – Paraphrasing -**Writing**-Writing a Paragraph – Paragraph Types **Vocabulary and Grammar**: If clause –– Articles – Sequence words – collocation– Conjunction.

UNIT	**TITLE**	**PERIODS**
V		**9**

Listening: TED /INK talks – Note taking **Speaking**: Participating in group discussion – Do's and Don'ts **Reading:** Reading reviews, advertisement – Newspaper/ magazines / Short stories - **Writing:** Process and Product description **Vocabulary and Grammar:**– Figures of Speech– Embedded sentences – WH Questions / Yes/No questions / Question tag

TOTAL PERIODS:	**45**

TEXT BOOKS:	
1.	V. Chellamal, Deepa Mary Francis, K.N Shoba, P.R Sujatha Priyadharsini, Veena Selvam English for science and Technology, Cambridge University Press and Assessment 2023
REFERENCE BOOKS:	
1.	Technical Communication – Principles And Practices By Meenakshi Raman & Sangeeta Sharma, Oxford Univ. Press, 2016, New Delhi.
2.	A Course Book on Technical English By Lakshmi Narayanan, Scitech Publications (India) Pvt. Ltd.
3.	English For Technical Communication (With CD) By Aysha Viswamohan, Mcgraw Hill Education,ISBN::0070264244.
4.	Effective Communication Skill, Kulbhusan Kumar, R S Salaria, Khanna Publishing House.
5.	Learning to Communicate – Dr. V. Chellammal, Allied Publishing House, New Delhi, 2003.

WEBSITES:	
1.	https://www.usingenglish.com
2.	http://grammarbook.com

JOURNALS:	
1.	National Council for Teachers of English
2.	https://www2.ncte.org/resources/journals/college-english/

EXTENSIVE READER:	
1.	Spencer Johnson, Who Moved My Cheese, Putnam Adult, 1998

COURSE DESIGNERS			
1.	Dr.Samuel Dawson	HoD	Department of English
2.	Dr.Usha Menon	Professor	Department of English
3.	Ms.Jean Ida	Assistant Professor	Department of English
4.	Ms.Moby	Assistant Professor	Department of English
5.	Ms.Surya S	Assistant Professor	Department of English
6.	Dr.Rudhra T S	Assistant Professor	Department of English

Recommended by Board of Studies	Date: 12.10.2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

		Periods per week	

231MAB101T	MATRICES AND CALCULUS	L	T	P	R	Credits
Regulation - R23	(Common to all branches of Engineering and Technology)	3	2	0	0	4

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:

Basic Knowledge of Matrices and Determinants, Differentiation

COURSE OBJECTIVES:

1.	To reduce quadratic form to canonical form of a matrix and identify its nature
2.	To analyze the convergence of infinite series
3.	To study the concept of evolutes and envelopes
4.	To find the extreme values for a function of two variables
5.	To solve Differential Equations using different techniques

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to:		Bloom"s level
CO1:	Reduce quadratic form to canonical form by orthogonal transformation and identify the nature of the quadratic form	K4
CO2:	Analyze the convergence of a given infinite series	K5
CO3:	Find evolute of a given curve and envelope of family of curves	K5
CO4:	Find the extreme of a linear function with two variables	K4
CO5:	Solve Ordinary differential equations with constant coefficients and variable coefficients	K6

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	-	-	-	-	-	-	-	-	1
CO2	2	2	1	-	-	-	-	-	-	-	-	1
CO3	2	1	1	-	-	-	-	-	-	-	-	1
CO4	2	2	1	-	-	-	-	-	-	-	-	1
CO5	2	1	1	-	-	-	-	-	-	-	-	1

3 – High : 2 - Medium : 1 – Low : „-„ - No correlation

UNIT	TITLE	PERIODS

I	MATRICES	9+3

Characteristic equation – Eigen values and Eigenvectors of a real matrix – Properties of Eigen- values andEigen-vectors, Cayley-Hamilton Theorem – statement and applications, Diagonalization of matrices by orthogonal transformation - Quadratic form – Reduction of a quadratic form to canonical form by Orthogonal transformation –Nature of quadratic forms

UNIT	TITLE	PERIODS
II	SEQUENCES AND SERIES	9+3

Sequences – Definition and examples, Series – Types of Convergence, Series of positive terms, Tests of convergence – Comparison test, Integral test and D'Alembert‟s ratio test, Alternating series – Leibnitz's test.

UNIT	TITLE	PERIODS
III	APPLICATIONS OF DIFFERENTIAL CALCULUS	9+3

Curvature, radius of curvature – Cartesian and parametric co-ordinates – Centre of curvature – Circle of curvature in Cartesian form, Evolutes, Envelopes, Evolute as envelope of normals.

UNIT	TITLE	PERIODS
IV	FUNCTIONS OF SEVERAL VARIABLES	9+3

Partial derivatives – Total derivative – Differentiation of implicit functions – Jacobian and its properties – Taylor's series for functions of two variables – Maxima and minima of functions of two variables –Lagrange's method of undetermined multipliers.

UNIT	TITLE	PERIODS
V	ORDINARY DIFFERENTIAL EQUATIONS	9+3

Solution of second and higher order linear differential equation with constants coefficients(Particular integrals involving f (x) = e^{mx}, sin mx, cos mx,x^m , e^{mx} f(x)). Euler's and Legendr's methods of solving Linear differential equations with variable coefficients, Method of variation of parameters.

TOTAL PERIODS:	60

TEXT BOOKS:

1.	Joel Hass, Christopher Heil and Maurice D. Weir "Thomas‟ Calculus", 14th Edition, Pearson.
2.	Grewal.B.S., "Higher Engineering Mathematics", Khanna Publishers, New Delhi, 44th Edition, 2018
3.	Work-Book on"" Matrices and Calculus", Chess Educational Publishers, prepared by Department of

REFERENCE BOOKS:

1	James Stewart, "Calculus: Early Transcendentals", Cengage Learning, 8th Edition, New Delhi, 2015

2	Anton. H, Bivens. I and Davis. S, "Calculus", Wiley, 10th Edition, 2016

WEBSITES:	
1.	www.pearsoned.co.in/georgebthomasjr
2.	www.cengage.com/international

COURSE DESIGNERS			
1.	Dr.S.Muthukumar	Professor	Department of Mathematics
2.	Dr.S.R.Ananthalakshmi	Associate Professor & HOD	Department of Mathematics
3.	Dr.K.S.Vidhyaa	Asst.Professor(Sr.G)	Department of Mathematics

Recommended by Board of Studies	Date: 12-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231PYB101T	ENGINEERING PHYSICS (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulations - R23		3	0	0	0	3

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:

Nil

COURSE OBJECTIVES:

1.	To impart knowledge on the basic principle of mechanics.
2.	To enable the students to gain knowledge on thermal physics.
3.	To explain the application of ultrasonics devices in engineering and medicine.
4.	To teach the description of various crystal structures and crystal defects for industrial applications.
5.	To learn the importance of laser and optical fibers for industry, telecommunication and medical applications.

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to:		Bloom's level
CO1:	Apply the basic principle of dynamics in torsional pendulum.	K3
CO2:	Evaluate the heat energy flow in thermal devices	K5
CO3:	Design ultrasonic devices for engineering and medical disciplines	K6
CO4:	Analyzethe crystal structures and crystal defects for industrial applications.	K4
CO5:	Select the appropriate laser and optical fibers for industry, telecommunication and medical applications.	K4

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	-	-	1
CO2	3	2	1	-	-	-	-	-	-	-	-	1
CO3	3	2	1	-	-	-	-	-	-	-	-	1
CO4	3	2	1	-	-	-	-	-	-	-	-	1
CO5	3	2	1	-	-	-	-	-	-	-	-	1

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT I	MECHANICS	9
Multiparticle dynamics: Center of mass (CM) – CM of continuous bodies – motion of the CM – kinetic energy of system of particles. Rotation of rigid bodies: Rotational kinematics – rotational kinetic energy and moment of inertia - theorems of M.I – moment of inertia of uniform rod, circular disc, solid cylinder – M.I of a diatomic molecule - torque – rotational dynamics of rigid bodies – conservation of angular momentum – gyroscope – Torsional pendulum.		
UNIT II	**THERMAL PHYSICS**	**9**
Fundamentals of thermal energy - Expansion joints - Bimetallic strips - Thermal conductivity, conductions in solids, Determination of thermal conductivity- Forbe's and Lee's disc method - Conduction through compound media – Thermal insulation – thermal shock resistance - Applications: Solar water heater- tempered glass- cryogenic materials.		
UNIT III	**SOUND WAVES AND VIBRATIONS**	**9**
Propagation, Intensity, Loudness of sound waves – Determination of absorption coefficient, Reverberation, Sabine's formula for reverberation time - Factors affecting acoustics of buildings and their remedies - Ultrasonic waves and properties, Methods of Ultrasonic production, Applications of Ultrasonic in engineering and medicine.		
UNIT IV	**CRYSTAL PHYSICS**	**9**
Single crystalline, polycrystalline and amorphous materials – single crystals: unit cell, crystal systems, Bravais lattices, directions and planes in a crystal, Miller indices – inter-planar distances - coordination number and packing factor for SC, BCC, FCC, HCP and diamond structures - crystal imperfections: point defects, line defects – Burger vectors, stacking faults.		
UNIT V	**APPLIED OPTICS**	**9**
Theory of laser - characteristics - Spontaneous and stimulated emission - Einstein's coefficients - population inversion –He-Ne laser, CO2 laser, Semiconductor laser – Basic applications of lasers in industry. Principle and propagation of light in optical fibre, Derivation for Numerical aperture and Acceptance angle - Types and losses of optical fibre - Fibre Optical Communication (Block diagram) – Fibre Optic Endoscope.		

TOTAL PERIODS:	**45**

TEXT BOOKS:	
1.	R. K. Gaur and S. L. Gupta, Engineering Physics, Dhanpat Rai Pub., 2018.
2.	Arthur Beiser, Shobhit Mahajan, S. Rai Choudhury, Concepts of Modern Physics, McGraw- Hill (Indian Edition), 2017.
3.	D.Kleppner and R.Kolenkow, An Introduction to Mechanics. McGraw Hill Education (Indian Edition), 2017.

REFERENCE BOOKS:	
1.	Jeff Sanny, Samuel J. Ling, and William Moebs, University Physics, Volume 1- 3, OpenStax, ISBN-13: 978-1-947172-15-9, 2023.
2.	D.Halliday, R.Resnick and J.Walker. Principles of Physics, Wiley (Indian Edition), 2015.
3.	Paul A. Tipler, Physic – Volume 1 & 2, CBS, (Indian Edition), 2004.
4.	D.K.Bhattacharya & T.Poonam, Engineering Physics, Oxford University Press, 2015.

5.	V. Rajendran, Engineering Physics, McGraw Hill Publication, 2017.

WEBSITES:	
1.	https://www.britannica.com/technology/laser/Laser-applications
2.	https://en.wikipedia.org/wiki/Crystal_structure

COURSE DESIGNERS			
1.	Dr. S. Nirmala	Assoc Professor & Head	Department of Physics
2.	Dr. G. Rajkumar	Professor	Department of Physics
3.	Dr. R. Sivakumar	Asst Professor	Department of Physics
4.	Dr. K. Raju	Asst Professor	Department of Physics

Recommended by Board of Studies	Date: 12-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231CYB101T	ENGINEERING CHEMISTRY (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		3	0	0	0	3

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:

Nil

COURSE OBJECTIVES:

1.	To explain water quality parameters and water treatment techniques
2.	To know about the polymers and polymer reinforced composites
3.	To gain knowledge about the types and applications of fuels
4.	To learn about various energy resources and storage devices
5.	To impart knowledge about the nanomaterials synthesis, properties and applications

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to:		Bloom's level
CO1:	Identify an appropriate water treatment technique for the given water sample	K4, Analyze
CO2:	Choose an appropriate method for polymer synthesis and for fabrication of plastics.	K3, Apply
CO3:	Select a suitable fuel for given application, based on the combustion characteristics of fuel.	K3, Apply
CO4:	Construct the device for electricity generation from available energy.	K3, Apply
CO5:	Synthesize the nanomaterials for a given application	K3, Apply

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	1	-	-	-	-	1
CO2	3	2	1	-	-	-	2	-	-	-	-	1
CO3	3	2	1	-	-	-	1	1	1	-	-	1
CO4	3	2	1	-	-	-	1	-	-	-	-	1
CO5	3	2	1	-	-	-	1	-	1	-	-	1

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS

I	WATER AND ITS TREATMENT	9

Introduction - types of impurities in water - water quality parameters: definition and significance of colour, odour, turbidity, pH, hardness, alkalinity, dissolved oxygen, chloride and fluoride - alkalinity - types and determination - hardness - types only - boiler feed water- requirements-boiler troubles - scale & sludge – disadvantages (wastage of fuels, decrease in efficiency, boiler explosion) - softening of hard water - external treatment process - demineralization and zeolite, internal treatment - boiler compounds (phosphate, calgon, carbonate and colloidal conditioning methods) – desalination of brackish water -reverse osmosis.

UNIT	TITLE	PERIODS
II	POLYMERS AND FRP COMPOSITES	9

Introduction- classification of polymers - natural and synthetic - thermoplastic and thermosetting, functionality – degree of polymerization,types - addition and condensation polymerization - free radical polymerization mechanism - preparation, properties and uses of PVC, Nylon 6,6, Teflon and Epoxy resin. Plastics - Compounding of plastics - moulding methods -injection, extrusion and compression – composites - definition, FRP composites

UNIT	TITLE	PERIODS
III	FUELS AND COMBUSTION	9

Classification - coal - proximate and ultimate analysis – carbonization - metallurgical coke - manufacture by Otto Hoffmann method - petroleum - refining - cracking -synthetic petrol by Bergius process – knocking in petrol and diesel engines- octane and cetane rating of fuels - synthesis - advantages and commercial application of alternate fuels (power alcohol, biodiesel and cryogenic fuel) - gaseous fuels - liquefied petroleum gases (LPG)- compressed natural gas (CNG) - Combustion of fuels: introduction - calorific value - higher & lower- theoretical calculation – flue gas analysis by Orsat method.

UNIT	TITLE	PERIODS
IV	ENERGY SOURCES AND STORAGE DEVICES	9

Energy - types - non-renewable energy - nuclear energy - fission and fusion reactions - differences between nuclear fission and fusion - nuclear chain reactions - light water nuclear reactor for power generation - breeder reactor - renewable energy - solar energy conversion - principle, working and applications of solar cells; recent developments in solar cell materials

Batteries - types of batteries - characteristics - construction and working of primary battery (dry cell) - secondary battery (lead acid battery and lithium-ion-battery), electric vehicles - working principles - fuel cells (H2-O2)

UNIT	TITLE	PERIODS
V	NANO CHEMISTRY AND GREEN NANOTECHNOLOGY	9

Introduction of Nano materials - distinction between nanoparticles, molecules and bulk materials – properties of nanomaterials - size dependent properties - synthesis – bottom up methods - precipitation, thermolysis (hydrothermal & solvothermal) - top down methods - electro-deposition, chemical vapour deposition, laser ablation – types of nanoparticles: nano cluster, nano-rod, nano-wire and nano-tube - carbon nano tube (synthesis and properties) - applications of nanoparticles.

Green nanotechnology - nanotechnology products as potential substitutes for harmful chemicals, environmental concerns with nanotechnology

TOTAL PERIODS:	45

TEXT BOOKS:

1.	Kannan P and Ravikrishnan A, - Engineering Chemistry, Sri Krishna, Hitech publishing Company Pvt. Ltd, 2023.
2.	Jain P.C. and Monika Jain, - Engineering Chemistry, Dhanpat Rai, Publishing Company (P) Ltd.,New Delhi, 2018.
3.	Sivasankar B., “Engineering Chemistry”, Tata McGraw-Hill Publishing Company Ltd, New Delhi, 2008.

REFERENCE BOOKS:	
1.	Dara S.S & S.S Umare, - A Text book of Engineering Chemistryǁ, S.Chand & Company Ltd., New Delhi, 2018.
2.	Palanna O.G, —Engineering Chemistryǁ, McGraw Hill Education (India)Pvt. Ltd, 2nd Edition, 2017.
3.	ShikhaAgarwal, "Engineering Chemistry-Fundamentals and Applications", Cambridge University Press, Delhi, 2nd Edition, 2019.

WEBSITES:	
1.	https://www.brainkart.com/subject/Engineering-Chemistry_264/
2.	https://www.poriyaan.in/paper/engineering-chemistry-4

JOURNALS:	
1.	Journal of water technology and treatment methods
2.	Journal of polymers and composites
3.	Fuel and combustion scientific and research journal
4.	Journal of renewable and sustainable energy
5.	Journal of nanomaterials

EXTENSIVE READER:	
1.	B. S. Murty, P. Shankar, Baldev Raj, B. B. Rath and James Murday, "Text book of nanoscience and nanotechnology", Universities Press-IIM Series in Metallurgy and Materials Science, 2018.
2.	Friedrich Emich, "Engineering Chemistry", Scientific International PVT, LTD, New Delhi, 2014.
3.	O.V. Roussak and H.D. Gesser, Applied Chemistry-A Text Book for Engineers and Technologists, Springer Science Business Media, New York, 2nd Edition, 2013.

COURSE DESIGNERS			
1.	Dr. C. Ravichandran	Professor	Chemistry
2.	Dr. V. Vanitha	Assistant Professor	Chemistry
3.	Dr. K. Saravanan	Associate Professor	Chemistry

Recommended by Board of Studies	Date: 12-10-2023	Syllabus version	1
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231GES104T	PROBLEMSOLVINGTHROUGHPYTHONPROGRAMMING (Common to CSE, CSE (CS), CSE (AIML), AI-DS, IT)	Periodsper week				Credits
		L	T	P	R	
Regulation-R23		2	0	0	0	2

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:
Nil

COURSEOBJECTIVES:	
1.	To introduce the basics of algorithmic problem solving
2.	To solve problems using Python conditionals and loops.
3.	To use Python function calls to solve problems.
4.	To impart Python data structures-lists, tuples, dictionaries to represent complex data
5.	To implement file operations using Python.

COURSEOUTCOMES(COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Solve problems using algorithms, flowchart and pseudocode.	Analyze
CO2:	Use python conditional and iteration statements for problem solving.	Apply
CO3:	Apply strings and user defined functions in python programming.	Apply
CO4:	Choose appropriate python data structures for real time applications.	Analyze
CO5:	DevelopPythoncodetomanipulatedatausingfileandexception-handling.	Create

MAPPINGOFCOURSEOUTCOMES(CO)WITHPROGRAMMEOUTCOME(PO)												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	2	2	1	0	0	0	0	0	2	1	1
CO2	1	2	1	1	0	0	0	0	0	1	0	1
CO3	2	2	1	1	0	0	0	0	0	1	0	1
CO4	2	2	1	1	0	0	0	0	0	1	0	1
CO5	1	3	2	2	0	2	1	0	0	1	1	1
3—High: 2-Medium:1—Low :„-„-No correlation												

UNIT	TITLE	PERIODS
I	ALGORITHMIC PROBLEM SOLVING	5
Fundamentals of Computing -Algorithms, building blocks of algorithms (statements, control flow, functions),notation (pseudo code, flow chart, programming language), algorithmic problem solving, simple strategies for developing algorithms (iteration, recursion).		
UNIT	TITLE	PERIODS
II	CONTROL FLOW STATEMENTS	7
Python interpreter, interactive mode and script mode; variables, expressions, statements; values and data types; Operators and Precedence of operators; Conditionals: conditional, alternative, chained conditional, nested conditional; Iterations: while, for, break, continue.		
UNIT	TITLE	PERIODS
III	FUNCTIONS AND STRINGS	6
Function definition and flow of execution, parameters and arguments; Fruitful functions, composition, recursion; Strings: string slices, immutability, Looping and counting, String methods.		
UNIT	TITLE	PERIODS
IV	LIST, TUPLE AND DICTIONARIES	8
Lists: list operations: list slices, , traversing, mutability, aliasing, list methods, list arguments, list comprehension; Tuples: Operations, tuple assignment; Dictionaries: Operations, functions and Looping.		
UNIT	TITLE	PERIODS
V	FILES, EXCEPTIONS	4
Files:Textfiles, reading and writing files,;Exceptions: handling exceptions, multiple exception blocks, finally block.		

TOTAL PERIODS:	30

TEXT BOOKS:	
1.	ReemaThareja—PythonProgrammingusingProblemsolvingApproach‖,OxfordUniversityPress.

REFERENCE BOOKS:	
1.	Robert Sedgewick, Kevin Wayne, Robert Dondero, Introduction to Programming in Python: An Inter- disciplinary Approach, Pearson India Education Services Pvt. Ltd., 2016.
2.	Paul Gries, Jennifer Campbell and Jason Montojo, Practical Programming: An Introduction to Computer Science using Python 3‖, Second edition, Pragmatic Programmers, LLC, 2013.
3.	Allen B. Downey, "Think Python: How to Think Like a Computer Scientist, Version 2.0.17 edition, Updated for Python 3,Shroff/O_ReillyPublishers, (http://greenteapress.com/wp/thinkpython/)

WEBSITES:	
1.	https://www.python.org
2.	https://www.learnpython.org
3.	https://www.w3schools.com/python/

JOURNALS:	
1.	Nil

EXTENSIVEREADER:	
1.	Nil

COURSEDESIGNERS			
1.	Dr.N.Ananthi	Professor & Head	Information Technology
2.	Dr.G.S.AnandhaMala	Professor & Head	Computer Science and Engineering
3.	Mrs.B.Chandra	Assistant Professor	Information Technology
4.	Dr.J.Deepa	Associate professor	Computer Science and Engineering
5	A.JebaSheela	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 20-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231GEH101T	HERITAGE OF TAMILS / தமிழர்மரபு (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		1	0	0	0	1

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:

Nil

COURSE OBJECTIVES:

1.	சங்க இலக்கியங்கள் முதல் நவீன இலக்கியங்கள் வரையிலான வளர்ச்சியைப் பற்றி அறிய வைத்தல் To know about the development from Sangam literature to modern literature
2.	பாறை ஓவியங்கள் முதல் நவீன சிற்பங்கள் வரையிலான தமிழர் கலைகள் பற்றி அறிந்துகொள்ள வைத்தல் Learn about Tamil art from rock paintings to modern sculptures
3.	தமிழர்களின் நாட்டுப்புற கலைகள் மற்றும் வீர விளையாட்டுகள் பற்றி தெரிந்துகொள்ள வைத்தல் To know about folk arts and heroic games of Tamils
4.	தமிழர்கள் போற்றிய அகம் மற்றும் புறக் கோட்பாடுகள் பற்றி தெரிந்து கொள்ள வைத்தல் To know about internal and external principles cherised by Tamils.
5.	இந்தியப் பண்பாட்டில் தமிழர்களின் பங்களிப்பு பற்றி அறிந்து கொள்ள வைத்தல் To know about the contribution of Tamils in Indian culture

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to: இந்தப்பாடத்தினைகற்றுமுடிக்கும்போது,மாணவர்கள்கீழ்க்கண்டவற்றைபற்றிஅறிந்திருப்பார்கள்:		Bloom's level
CO1:	சங்க இலக்கியங்கள் முதல் நவீன இலக்கியங்கள் வரையிலான வளர்ச்சியைப் பற்றி. On the Development from Sangam Literature to Modern Literature	K2
CO2:	பாறை ஓவியங்கள் முதல் நவீன சிற்பங்கள் வரையிலான தமிழர் கலைகள் பற்றி. About Tamil arts from rock paintings to modern sculptures.	K2
CO3:	தமிழர்களின் நாட்டுப்புற கலைகள் மற்றும் வீர விளையாட்டுகள் பற்றி. About folk arts and heroic games of Tamils	K2
CO4:	தமிழர்கள் போற்றிய அகம் மற்றும் புறக் கோட்பாடுகள் பற்றி. About internal and external principles cherished by Tamils.	K2
CO5:	இந்தியப் பண்பாட்டில் தமிழர்களின் பங்களிப்பு பற்றி. About the contribution of Tamils in Indian culture.	K2

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	LANGUAGE AND LITERATURE/மொழிமற்றும்இலக்கியம்	3

Language Families in India - Dravidian Languages – Tamil as a Classical Language - Classical Literature in Tamil – Secular Nature of Sangam Literature – Distributive Justice in Sangam Literature - Management Principles in Thirukural - Tamil Epics and Impact of Buddhism & Jainism in Tamil Land - Bakthi Literature Azhwars and Nayanmars - Forms of minor Poetry - Development of Modern literature in Tamil - Contribution of Bharathiyar and Bharathidhasan.

இந்தியமொழிக்குடும்பங்கள்–திராவிடமொழிகள்–தமிழ்ஒருசெம்மொழி–தமிழ்செவ்விலக்கியங்கள் – சங்கஇலக்கியத்தின்சமயச்சார்பற்றதன்மை-சங்கஇலக்கியத்தில்பகிர்தல்அறம்– திருக்குறளில்மேலாண்மைக்கருத்துக்கள்-தமிழ்க்காப்பியங்கள், தமிழகத்தில்சமணபௌத்தசமயங்களின்தாக்கம்பக்திஇலக்கியம், ஆழ்வார்கள்மற்றும்நாயன்மார்கள் - சிற்றிலக்கியங்கள் - தமிழில்நவீனஇலக்கியத்தின்வளர்ச்சி– தமிழ்இலக்கியவளர்ச்சியில்பாரதியார்மற்றும்பாரதிதாசன்ஆகியோரின்பங்களிப்பு.

UNIT	TITLE	PERIODS
II	HERITAGE - ROCK ART PAINTINGS TO MODERN ART – SCULPTURE/ மரபு– பாறைஓவியங்கள்முதல்நவீனஓவியங்கள்வரை- சிற்பக்கலை	3

Hero stone to modern sculpture - Bronze icons - Tribes and their handicrafts - Art of temple car making - - Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari, Making of musical instruments - Mridhangam, Parai, Veenai, Yazh and Nadhaswaram - Role of Temples in Social and Economic Life of Tamils.

நடுகல்முதல்நவீனசிற்பங்கள்வரை–ஐம்பொன்சிலைகள்- பழங்குடியினர்மற்றும்அவர்கள்தயாரிக்கும்கைவினைப்பொருட்கள், பொம்மைகள்– தேர்செய்யும்கலை– சுடுமண்சிற்பங்கள்நாட்டுப்புறத்தெய்வங்கள்குமரிமுனையில்திருவள்ளுவர்சிலை - இசைக்கருவிகள் - மிருதங்கம், பறை, வீணை, யாழ். நாதஸ்வரம்– தமிழர்களின்சமூகபொருளாதாரவாழ்வில்கோவில்களின்பங்கு.

UNIT	TITLE	PERIODS
III	FOLK AND MARTIAL ARTS / நாட்டுப்புறக்கலைகள்மற்றும்வீரவிளையாட்டுகள்	3
Therukoothu, Karagattam, Villu Pattu, Kaniyan Koothu, Oyillattam, Leather puppetry, Silambattam, Valari, Tiger dance - Sports and Games of Tamils.		
தெருக்கூத்து, கரகாட்டம், வில்லுப்பாட்டு, கணியான்கூத்து, ஒயிலாட்டம், தோல்பாவைக்சிலம்பாட்டம், வளரி, புலியாட்டம், தமிழர்களின்விளையாட்டுகள்.		

UNIT	TITLE	PERIODS
IV	THINAI CONCEPT OF TAMILS /தமிழர்களின்திணைக்கோட்பாடுகள்	3
Flora and Fauna of Tamils & Aham and Puram Concept from Tholkappiyam and Sangam Literature - Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age - Export and Import during Sangam Age - Overseas Conquest of Cholas.		
தமிழகத்தின்தாவரங்களும், விலங்குகளும்தொல்காப்பியம்மற்றும்சங்கஇலக்கியத்தில்அகம்மற்றும்பறக்கோட்பாடுகள்தமிழர்கள்போற்றியஅறக்கோட்பாடு–சங்ககாலத்தில்தமிழகத்தில்எழுத்தறிவும், கல்வியும்–சங்ககாலநகரங்களும்துறைமுகங்களும்–சங்ககாலத்தில்ஏற்றுமதிமற்றும்இறக்குமதிகடல்கடந்தநாடுகளில்சோழர்களின்வெற்றி.		

UNIT	TITLE	PERIODS
V	CONTRIBUTION OF TAMILS TO INDIAN NATIONAL MOVEMENT AND INDIAN CULTURE /இந்தியதேசியஇயக்கம்மற்றும்இந்தியபண்பாட்டிற்குத்தமிழர்களின்பங்களிப்பு	3
Contribution of Tamils to Indian Freedom Struggle - The Cultural Influence of Tamils over the other parts of India – Self-Respect Movement - Role of Siddha Medicine in Indigenous Systems of Medicine – Inscriptions & Manuscripts – Print History of Tamil Books.		
இந்தியவிடுதலைப்போரில்தமிழர்களின்பங்குஇந்தியாவின்பிறப்பகுதிகளில்தமிழ்ப்பண்பாட்டின்தாக்கம்–சுயமரியாதைஇயக்கம்–இந்தியமருத்துவத்தில், சித்தமருத்துவத்தின்பங்கு-கல்வெட்டுகள், கையெழுத்துப்படிகள்–தமிழ்ப்புத்தகங்களின்அச்சுவரலாறு.		

TOTAL PERIODS:	15

TEXT BOOKS CUM REFERENCE BOOKS:	
1.	தமிழக வாலாறு மக்களும் பண்பாடும் கே.கே. பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2.	கணினித்தமிழ் - முனைவர்இல. சுந்தரம். (விகடன்பிரசுரம்).
3.	வைகைநதிக்கரையில்சங்ககாலநகரநாகரிகம் (தொல்லியல்துறைவெளியீடு)
4.	பொருறை - ஆற்றங்கரைநாகரிகம், தொல்லியல்துறைவெளியீடு)
5.	Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6.	Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
7.	Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8.	The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
9.	Keeladi - 'Sangam City C ivilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10.	Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by: The Author)
11.	Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational

	Services Corporation, Tamil Nadu)
12.	Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

COURSE DESIGNERS	
1.	Anna University, Chennai

Recommended by Board of Studies	Date:	Syllabus version	-
Approved by the Academic Council	Date:	Meeting No.	6



231CYB011L	CHEMISTRY LABORATORY (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		0	0	2	0	1

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	60 %	40 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To acquire practical skills in the determination of water quality parameters.
2.	To impart the students with the determination of the molecular weight of the polymer by using a viscometer.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Determine the water quality parameters like hardness and DO content.	K3, Apply
CO2:	Calculate the molecular weight and classify the polymers.	K3, Apply
CO3:	Estimate the strength of acids using instrumental techniques.	K3, Apply
CO4:	Analyze the alkalinity parameters of water.	K4, Analyze

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	1	-	-	-	-	1
CO2	3	2	-	-	-	-	1	-	-	-	-	1
CO3	3	2	-	-	-	-	1	-	-	-	-	1
CO4	3	2	-	-	-	-	1	-	-	-	-	1

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

List of experiments: (Any Five Experiments)

1. Determination of alkalinity in the given water sample
2. Estimation of total, temporary & permanent hardness of water by EDTA method
3. Determination of chloride content of water sample by Argentometric method
4. Find the molecular weight of PVA using Ostwald viscometer
5. Determine the strength of given strong acid using pH meter
6. Evaluate the strength of acids in a mixture by conductometric titration
7. Estimation of Fe2+ by Potentiometric titration
8. Determination of BOD and COD in water sample

Content beyond the syllabus: (Any one)

9. Demonstrate the of DO Content of water sample by Wrinkler method
10. Estimation of Iron content of water sample using spectrophotometer (1,10 – Phenanthroline/ thiocyanate method)

TOTAL PERIODS:	**30**

TEXT BOOKS:	
1.	Dr. C. Ravichandran, "Engineering Chemistry Laboratory" Global publications, India, (2020)
2.	Furniss B.S. Hannaford A.J, Smith P.W.G and Tatchel A.R., "Vogel's Textbook of practical organic chemistry, LBS Singapore (1994)

COURSE DESIGNERS			
1.	Dr. C. Ravichandran	Professor	Chemistry
2.	Dr. R. Anithadevi	Assistant Professor	Chemistry
3.	Dr. V.Vanitha	Assistant Professor	Chemistry

Recommended by Board of Studies	Date: 12.10.2023	Syllabus version	1.0
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231GES011L	**BASIC WORKSHOP PRACTICE**	Periods per week				Credits
		L	T	P	R	
Regulation - R23	**(Common to all branches of Engineering and Technology)**	**0**	**0**	**2**	**0**	**1**

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	60 %	40 %		-	45%	50%

PREREQUISITES:

Nil

COURSE OBJECTIVES:

1.	To provide exposure to the students with the concepts involved in product realization by carrying out basic workshop exercises.
2.	Hands-on practice with exercises and assembly leading to realization of a new product in a group

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to:		Bloom's level
CO1:	Prepare the pipe connections using pipes and joints	Apply
CO2:	Prepare welding joints using arc welding.	Apply
CO3:	Make carpentry joints.	Apply
CO4:	Prepare objects using sheet metal.	Apply

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	1	-	1	3	1	2	2	-	1	3
CO2	3	-	1	-	1	3	1	2	2	-	1	3
CO3	3	-	1	-	1	3	1	2	2	-	1	3
CO4	3	-	1	-	1	3	1	2	2	-	1	3

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

LIST OF EXPERIMENTS

PART A - CIVIL ENGINEERING PRACTICE

Plumbing Works:

1. Study of pipeline joints, its location and functions: Valves, Taps, Couplings, Unions, Reducers, Elbows in house hold fittings.
2. Connection of PVC pipes
3. Basic pipe connections involving the fitting like Valves, Taps and Bends

Carpentry works:

4. Study of the joints in Roofs, Doors, Windows and Furniture.
5. Cross Lap joint
6. Dove Tail Joint

PART B - MECHANICAL ENGINEERING PRACTICE

Welding:

7. Introduction to Welding
8. Arc welding of Butt joints, Tap joints and Tee joints.
9. Gas welding practice

Sheet metal work:

10. Introduction to sheet metal
11. Rectangular tray making
12. Funnel making

Group Task: Model house creation using plumbing and carpentry tools.

TOTAL PERIODS:	**30**

TEXT BOOKS:

1.	Hajara Choudhary, Bose S K, Elements of Workshop Technology Vol I and II, Asia Publishing House.

REFERENCE BOOKS:

1.	Rao P N, Manufacturing Technology and Foundry, Forming and Welding, Tata McGrawHill publishing Company
2.	Parmar R S, Welding Process and Technology, Khanna Publisher.
3.	Jain R K, Production Technology, Khanna Publisher

WEBSITES:

1.	https://pdf.usaid.gov/pdf_docs/PNAAM100.pdf
2.	https://content.kopykitab.com/ebooks/2016/06/7478/sample/sample_7478.pdf

JOURNALS:	
1.	https://www.irjet.net/archives/V6/i4/IRJET-V6I4824.pdf
2.	https://www.sciencedirect.com/science/article/pii/S2352710223012688

COURSE DESIGNERS			
1.	Dr. M. Vetrivel Sezhian	Professor	Mechanical Engineering
2.	Dr. A. Praveen Kumar	Assistant Professor	Mechanical Engineering

Recommended by Board of Studies	Date: 16-11-2023	Syllabus version	
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231GES013L	BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY	Periods per week				Credits
		L	T	P	R	
Regulation - R23	(Common to AI&DS, Automobile, BioTech, Civil, CSBS, CSD, CSE (AIML), CSE (CS), Mechanical, IT)	0	0	4	0	2

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	60 %	40 %		-	45%	50%

PREREQUISITES: NIL

COURSE OBJECTIVES:

1.	To train the students in residential wiring.
2.	To gain practical knowledge in performance characteristics ofanalog devices
3.	To acquire exposure in combinational circuits

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to:		Bloom's level
CO1:	Carry out the simple electrical wiring of a residential building.	Apply
CO2:	Measure the electrical quantities using Energy meter and Megger	Apply
CO3:	Calculate the voltage and current parameters of given circuits using Kirchhoff's Laws.	Apply
CO4:	Analyze the characteristics of analog devices.	Analyze
CO5:	Construct combinational circuits with logic gates.	Apply

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	1	-	-	1
CO2	3	2	2	-	-	-	-	-	1	-	-	1
CO3	3	2	2	-	-	-	-	-	1	-	-	1
CO4	3	2	2	-	-	-	-	-	1	-	-	1
CO5	3	2	2	-	-	-	-	-	1	-	-	1

3 – High : 2 - Medium : 1 – Low : '-' - No correlation	
LIST OF EXPERIMENTS	
1.	Fluorescent Lamp wiring.
2.	Staircase wiring.
3.	Fabricate and test a PCB layout for a given circuit.
4.	Study of Single-Phase Energy meter.
5.	Study of earth resistance measurement using Megger.
6.	Verification of Kirchhoff's Laws.
7.	Characteristics of PN and Zener Diodes.
8.	Characteristics of BJT.
9.	Half wave and Full Wave rectifiers.
10.	Study of Logic Gates.
11.	Implementation of Binary Adder and Subtractor.
12.	Characteristics of the LVDT.

TOTAL PERIODS:	**60**

REFERENCE BOOKS:	
1.	Mittle V.N, Arvind Mittal, "Basic Electrical Engineering", Tata Mc Graw Hill (India), Second Edition, 2013.
2.	Sedha R.S., "A Text Book of Applied Electronics", S. Chand & Co., 2014

COURSE DESIGNERS			
1.	Dr. M. Sujatha	Assistant Professor	EEE
2.	Mrs. D. Chandrakala	Assistant Professor	EEE
3.	Mrs. K. A. Indu Sailaja	Assistant Professor	EEE

Recommended by Board of Studies	Date: 30.10.2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231GES114L	PYTHONPROGRAMMINGLABORATORY	Periods per week				Credits
		L	T	P	R	
Regulation-R23	(Common to CSE, CSE (CS), CSE (AIML), AI-DS, IT)	**0**	**0**	**3**	**1**	**2**

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	60 %	40 %		-	45%	50%

COURSEOBJECTIVES:	
1.	To impart knowledge on problem solving approaches.
2.	To experiment with programming constructs in Python
3.	To practice various computing strategies for Python-based solutions to real world problems
4.	To use Python data structures-lists, tuples, dictionaries.
5.	To train in building simple project using python.

COURSEOUTCOMES:		Bloom's level
Upon completion of this course, student will be able to:		Analyze
CO1:	Develop algorithmic solutions to simple computational problems	Apply
CO2:	Implement programs in Python using conditionals and loops for solving problems	Apply
CO3:	Deploy functions to decompose a Python program	Apply
CO4:	Process compound data using Python data structures.	Apply
CO5:	Design python programs for file handling and exception handling.	Apply
CO6:	Create GUI application for user defined requirement.	Create

MAPPINGOFCOURSEOUTCOMES(CO)WITHPROGRAMMEOUTCOME(PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	1	2	2	1	1	-	-	-	-	2	1	2
CO2	1	2	1	1	1	-	-	-	1	2	1	2
CO3	2	2	1	1	1	-	-	-	1	2	1	2
CO4	2	2	1	1	1	-	-	-	1	2	1	2
CO5	1	3	2	2	1	-	-	-	1	2	1	2
CO6	2	2	2	2	2	-	1	2	3	3	2	3
3—High:2-Medium:1—Low :„-„-No correlation												

LIST OF EXPERIMENTS	
1.	Identification and solving of simple real life or scientific or technical problems, and developing flow charts, algorithms and pseudocode for the same. (Electricity Billing / Retail shop billing/ Sin series/ weight of a motorbike/ Weight of a steel bar/ compute Electrical Current in Three Phase AC Circuit, etc.)
2.	Python programming using simple statements and expressions (exchange the values of two variables/ circulate the values of n variables/ distance between two points).
3.	Scientific problems using Conditionals and Iterative loops.(Number series/Number Patterns/pyramid pattern)
4.	Implementing real-time/technical applications using Lists, Tuples. (Items present in a library/Components of a car/ Materials required for construction of a building –operations of list & tuples)
5.	Implementingreal-time/technicalapplicationsusingSets,Dictionaries.(Language/componentsofan automobile / Elements of a civil structure etc.- operations of Sets & Dictionaries)
6.	Implementing programs using Functions. (Factorial/ largest number in a list/ area of shape)
7.	Implementing programs using Strings.(reverse/palindrome/character count/ replacing characters)
8.	Implementing real-time / technical applications using File handling.(copy from one file to another, word count, longest word)
9.	Implementing real-time/technical applications using Exception handling. (divide by zero error / voter"s age validity / student mark range validation)
10.	ExploringPygametooltoDevelopagameactivitylikebouncingball,carrace etc.
11.	Mini Project (any ONE): Design GUI for • Airline reservation system • Feedback system • Employee management system • Student management system • Banking system
TOTALPERIODS:	**60**

EXPERIMENTSBEYONDTHESYLLABUS(Anyone)	
1.	Implementing programs using written modules and Python Standard Libraries-pandas,numpy.
2.	Implementing programs using written modules and Python Standard Libraries-Matplotlib,spicy

COURSEDESIGNERS			
1.	Dr.N.Ananthi	Professor &Head	Information Technology
2.	Dr.G.S.Anandha Mala	Professor & Head	Computer Science and Engineering
3.	Mrs.B.Chandra	Assistant Professor	Information Technology
4.	Dr.J.Deepa	Associate professor	Computer Science and Engineering
5.	A.JebaSheela	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 20-10-2023	SyllabusVersion	1
ApprovedbytheAcademicCouncil	Date: 24-01-2024	MeetingNo.	6



SEMESTER II

231LEH201T	PROFESSIONAL COMMUNICATION (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To develop the basic writing skills of the First year Engineering students.
2.	To help learners develop their listening skills, which will, enable them to listen to lectures and enhance their ability to comprehend by asking questions and seeking clarification.
3.	To help learners develop their speaking skills and help them to speak fluently.
4.	To inculcate reading habit and to develop effective reading skills.
5.	To help students improve their active and passive vocabulary.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Listen, Understand and Respond to others in different situations	K2
CO2:	Speak correctly and fluently in different situations using appropriate communication strategies.	K4
CO3:	Read and Comprehend a range of texts adopting different reading skills.	K3
CO4:	Write with clarity in simple, apt and flawless language with coherence and cohesion.	K6
CO5:	Use their communicative competency with purpose and clarity in the context of Science and Technology	K6

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	-	-	-	-	-	-	-	-	1	1	-	-
CO2	-	-	-	-	-	-	-	-	1	2	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	1	-	1
CO6	-	-	-	-	-	-	-	-	1	-	-	-
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

UNIT	TITLE	PERIODS
I	Listening: Listening to talks and complete information gap exercises. (Comprehension).Speaking: Introduction to speaking mechanics. Reading: Prescribed non-detail text .Writing: Communication- Process/ forms /barriers. Vocabulary and Grammar: Purpose and function- Active/Passive voice/Impersonal passive voice - Numerical adjectives - Words used as different parts of speech – Contextual meaning of words	**9**
UNIT	**TITLE**	**PERIODS**
II	Listening: Listening to documentaries and interpret Speaking: Speaking an advertisements/ gadget reviews user manuals Reading: Prescribed non-detail text - Reading longer technical texts Writing: Interpreting Chart/ graphs – letter writing (placing order/ escalation/ complaint) Vocabulary and Grammar: Transformation of sentences –Simple, Complex, and Compound sentences- Modal verb –Infinity and Gerund – Error correction.	**9**
UNIT	**TITLE**	**PERIODS**
III	Listening: Listening to technical talk and making notes. Speaking: Describing a process. Reading: Prescribed non-detail text- reading editorials and opinion blogs. Writing: Essay writing (descriptive/ argumentative/ analytical/ narrative).Vocabulary and Grammar: Words often misspelt –Synonyms and antonyms – Modifiers – Direct / Indirect.	**9**
UNIT	**TITLE**	**PERIODS**
IV	Listening: Listening to Online interviews and discussion. Speaking: Small Talk (any random topics Reading: Prescribed non-detail text .Writing: Report Writing (accident/ industrial / Feasibility/ project) Vocabulary and Grammar: Extended definition – cause and effect expressions –Verbal analogies- Compound nouns.	**9**
UNIT	**TITLE**	**PERIODS**
V	Listening: Listening for taking notes and seeking clarifications. Speaking: Group Discussion- Etiquette/ Do's and Dont's. Reading:Prescribed non-detail text .Writing: Cover letter and Resume writing – Minutes of Meeting Vocabulary and Grammar: Collective noun - Relative clause – Redundancies.	**9**

TOTAL PERIODS:	**45**

TEXT BOOKS:	
1.	V. Chellamal, Deepa Mary Francis, K.N Shoba, P.R Sujatha Priyadharsini, Veena Selvam English for science and Technology, Cambridge University Press and Assessment 2023
REFERENCE BOOKS:	
1.	Dutt P. Kiranmai and RajeevanGeeta. Basic Communication Skills, Foundation Books: 2013
2.	Bailey, Stephen. Academic Writing: A practical guide for students. New York: Rutledge,2011.
3.	Means,L. Thomas and Elaine Langlois. English & Communication for Colleges. Cengage Learning ,USA: 2007.

WEBSITES:	
1.	https://www.usingenglish.com
2.	http://grammarbook.com

JOURNALS:	
1.	National Council for Teachers of English
2.	https://www2.ncte.org/resources/journals/college-english/

EXTENSIVE READER:	
1.	Spencer Johnson, Who Moved My Cheese, Putnam Adult, 1998

COURSE DESIGNERS			
1.	Dr Usha Menon	Professor	Department of English
2.	Dr Sam Dawson	Professor and Head	Department of English
3.	Ms Jean Ida	Assistant Professor	Department of English
4.	Ms Moby	Assistant Professor	Department of English
5	Ms Surya S	Assistant Professor	Department of English
6	Dr Rudhra T S	Assistant Professor	Department of English

Recommended by Board of Studies	Date: 12-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231MAB201T	ADVANCED CALCULUS AND COMPLEX ANALYSIS (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		3	2	0	0	4

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:

Basic Knowledge of Complex Variables, Vector Algebra and Integration

COURSE OBJECTIVES:

1.	To compute area of closed surface and volume of solids using multiple integrals
2.	To evaluate Line, Surface and Volume integrals
3.	To find the Laplace Transforms and inverse transforms for standard functions
4.	To apply C-R equations in the construction of Analytic Functions
5.	To study the methods of Complex Integration, finding Taylor's and Laurent's Series expansions

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to:		Bloom's level
CO1:	Find the area enclosed by simple closed curve using double integral and volume of solid	K6
CO2:	Verify Green's, Stoke's and Gauss divergence theorems	K6
CO3:	Obtain Laplace Transforms and inverse transforms of standard functions	K5
CO4:	Solve problems in Analytic functions and construct analytic functions by using C-R	K5
CO5:	Identify a suitable method of complex integration for evaluating certain indefinite integrals	K4

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	2	-	-	-	-	-	-	-	-	1
CO2	2	1	2	-	-	-	-	-	-	-	-	1
CO3	2	2	1	-	-	-	-	-	-	-	-	1
CO4	2	1	2	-	-	-	-	-	-	-	-	1
CO5	2	2	1	-	-	-	-	-	-	-	-	1

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	**MULTIPLE INTEGRALS**	9+3

Double integrals in Cartesian and polar coordinates – Change of order of integration, Area enclosed by planecurves – Change of variables in double integrals, Triple integrals

UNIT	TITLE	PERIODS
II	**VECTOR CALCULUS**	9+3

Gradient and directional derivative – Divergence and curl – Irrotational and Solenoidal vector fields, Line integral over a plane curve, Surface integral – Area of a curved surface, Volume integral, Green's, Gauss divergence and Stoke's theorems – Verification and application in evaluating line, surface and volume integrals (Cubes andCuboids only)

UNIT	TITLE	PERIODS
III	**LAPLACE TRANSFORM**	9+3

Definition, properties, existence conditions – Transforms of elementary functions – Transform of unit step function and unit impulse function, shifting theorems, transforms of derivatives and integrals, Initial and final value theorems, Periodic functions, Inverse transforms – Convolution theorem-, Solving linear second order ordinary differentialequations with constant coefficients using Laplace transforms

UNIT	TITLE	PERIODS
IV	**ANALYTIC FUNCTIONS**	9+3

Analytic functions – necessary and sufficient conditions, Cauchy-Riemann equations in Cartesian and polar form – Properties – harmonic functions, Construction of analytic function by Milne Thomson method, conformal mapping – some standard transformations w = z + c, cz, 1/z, bilinear transformation.

UNIT	TITLE	PERIODS
V	**COMPLEX INTEGRATION**	9+3

Line integral – Cauchy's integral theorem – Cauchy's integral formula, Taylor's and Laurent's series, Singularities – Residues – Residue theorem – Application of residue theorem for evaluation of real integrals – Use of circularcontour and semicircular contour (except the poles on the real axis). line, surface and volume integrals.

TOTAL PERIODS:	**60**

TEXT BOOKS:	
1.	Joel Hass, Christopher Heil and Maurice D. Weir “Thomas’ Calculus”, 14th Edition, Pearson.
2.	Grewal.B.S., “Higher Engineering Mathematics”, Khanna Publishers, New Delhi, 44th Edition, 2018
3.	Workbook on”Advanced calculus and Complex Analysis”,Chess Publications,Prepared by Department of Mathematics

REFERENCE BOOKS:	
1.	Bali N P, Manish Goyal, “A Textbook of Engineering Mathematics”, Ninth Edition, Laxmi Publications Pvt Ltd,
2.	James Stewart, "Calculus: Early Transcendentals", Cengage Learning, 8th Edition, New Delhi, 2015
3.	Kreyszig.E, "Advanced Engineering Mathematics", John Wiley and Sons,
4.	T Veerarajan,” Engineering Mathematics II”, McGraw Hill (India) Private Limited, Chennai

WEBSITES:	
1.	www.pearsoned.co.in/georgebthomasjr
2.	www.webassign.net
3.	

COURSE DESIGNERS			
1.	Dr.S.Muthukumar	Professor	Department of Mathematics
2.	Dr.S.R.Ananthalakshmi	HOD/Associate Professor	Department of Mathematics
3.	Dr.K.S.Vidhyaa	Assistant Professor(Sr.G)	Department of Mathematics

Recommended by Board of Studies	Date:12.10.2023	Syllabus version	V.1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231PYS202T	PHYSICS FOR INFORMATION SCIENCE (Common to CSE, IT, CS, AIML and AI & DS)	Periods per week				Credits
		L	T	P	R	
Regulations - R23		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To teach the principles of quantum mechanics and transport phenomena for semiconducting materials.
2.	To explain the functioning of optical materials for optoelectronics.
3.	To introduce the basic principles of quantum mechanics to one dimensional motion of particles.
4.	To enable the students to gain knowledge on applied quantum mechanics to form energy bands.
5.	To teach the basics of quantum structures and their applications and quantum computing.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Apply the principles of quantum mechanics and transport phenomena for semiconducting materials.	K3
CO2:	Design the optical materials and devices for optoelectronics	K6
CO3:	Apply the principles of quantum mechanics to one dimensional motion of particles.	K3
CO4:	Analyze the quantum mechanical principles towards the formation of energy bands.	K4
CO5:	Recommend the nano devices and nano materials for quantum computing	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	2	1	-	-	-	-	-	-	-	-	1
CO2	3	2	1	-	-	-	-	-	-	-	-	1
CO3	3	2	1	-	-	-	-	-	-	-	-	1
CO4	3	2	1	-	-	-	-	-	-	-	-	1
CO5	3	2	1	-	-	-	-	-	-	-	-	1
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

UNIT	TITLE	PERIODS
I	**SEMICONDUCTOR PHYSICS**	**9**

Intrinsic Semiconductors – Energy band diagram – direct and indirect band gap semiconductors – Carrier concentration in intrinsic semiconductors – extrinsic semiconductors - Carrier concentration in N-type & P-type semiconductors (Qualitative study) – Variation of carrier concentration with temperature – variation of Fermi level with temperature and impurity concentration – Carrier transport in Semiconductor: random motion, drift, mobility and diffusion – Hall effect and devices – Ohmic contacts – Schottky diode.

UNIT	TITLE	PERIODS
II	**LIGHT - SEMICONDUCTOR INTERACTION**	**9**

Classification of optical materials – carrier generation and recombination processes - Absorption emission and scattering of light in metals, insulators and semiconductors (concepts only) - photo current in a PIN diode – solar cell - LED – Organic LED – Laser diodes – Optical data storage techniques.

UNIT	TITLE	PERIODS
III	**BASIC QUANTUM MECHANICS**	**9**

Inadequacies of Classical Mechanics – Black body radiation - Planck's theory of radiation - Dual nature of electromagnetic radiation – De Broglie hypothesis for matter waves – Heisenberg's uncertainty principle – Schrodinger's time dependent and independent wave equation, significance of wave function - Born interpretation - Particle confinement in 1D box.

UNIT	TITLE	PERIODS
IV	**APPLIED QUANTUM MECHANICS**	**9**

The harmonic oscillator - Barrier penetration and quantum tunneling – Scanning Tunneling Microscope (STM) - Resonant diode - Finite potential wells - Bloch's theorem for particles in a periodic potential – Basics of Kronig - Penney model and origin of energy bands.

UNIT	TITLE	PERIODS
V	NANODEVICES AND QUANTUM COMPUTING	9
Introduction - quantum confinement – quantum structures: quantum wells, wires and dots – band gap of nanomaterials. Tunneling – Single electron phenomena: Coulomb blockade – resonant tunneling diode – single electron transistor – quantum system for information processing - quantum states – classical bits – quantum bits or qubits – CNOT gate - multiple qubits – Bloch sphere – quantum gates – advantage of quantum computing over classical computing.		

TOTAL PERIODS:	45

TEXT BOOKS:	
1.	Jasprit Singh, “Semiconductor Devices: Basic Principles”, Wiley (Indian Edition), 2007.
2.	S.O. Kasap. Principles of Electronic Materials and Devices, McGraw-Hill Education (Indian Edition), 2020.
3.	Parag K. Lala, Quantum Computing: A Beginner's Introduction, McGraw-Hill Education (Indian Edition), 2020.

REFERENCE BOOKS:	
1.	Charles Kittel, Introduction to Solid State Physics, Wiley India Edition, 2019.
2.	Y. B. Band and Y. Avishai, Quantum Mechanics with Applications to Nanotechnology and Information Science, Academic Press, 2013.
3.	B. Rogers, J.Adams and S.Pennathur, Nanotechnology: Understanding Small Systems, CRC Press, 2014.
4.	Nouredine Zettili, Quantum Mechanics Concepts and Applications, 2nd Edition, Wiley, 2009.
5.	V.Rajendran, Materials Science, McGraw Hill Education (India) Private Ltd., 2017.
6.	G. Aruldhas, Quantum Mechanics, PHI Learning, 2008.

WEBSITES:	
1.	https://nptel.ac.in/courses/115102025

COURSE DESIGNERS			
1.	Dr. S. Nirmala	Associate Professor & Head	Department of Physics
2.	Dr. G. Rajkumar	Professor	Department of Physics
3.	Dr. R. Sivakumar	Assistant Professor	Department of Physics
4.	Dr. K. Raju	Assistant Professor	Department of Physics

Recommended by Board of Studies	Date: 12-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231GEB201T	ENVIRONMENTAL SCIENCE FOR COMPUTING SCIENCES (Common to CSE, IT, CS, AIML and AI & DS)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		2	0	0	0	2

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES: NIL

COURSE OBJECTIVES:	
1.	To provide a basic understanding about the Ecology and Environmental agreements.
2.	To explain various natural resources and its conservation.
3.	To apply engineering knowledge to solve environmental pollution and its global issues.
4.	To analyze the concepts of various geospatial technology.
5.	To apply the geospatial technology in environmental aspects.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Analyze various threats to biodiversity, its conservation and environmental agreements.	K4, Analyze
CO2:	Select a suitable method to conserve natural resources for sustainable development.	K4, Analyze
CO3:	Apply necessary steps for pollution prevention, global issues and disaster management.	K3, Apply
CO4:	Analyze the various tools used in the geospatial technology	K4, Analyze
CO5:	Apply the geospatial technology in environmental studies	K3, Apply

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	1	-	1	1	-	-	-	-	1
CO2	2	2	-	1	-	1	2	-	1	-	-	1
CO3	2	2	-	1	-	1	1	1	1	-	-	1
CO4	3	2	2	2	2	1	1	-	-	-	-	1
CO5	3	2	2	2	2	1	2	-	-	-	-	1

3 – High : 2 - Medium : 1– Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	**Ecological biodiversity and its agreement**	6

Definition and scope of an environment – ecosystem (biotic and abiotic components) – ecological succession – food chain, food web – values of biodiversity – threats to biodiversity – conservation of biodiversity: In-situ and Ex-situ. Environmental Conventions and Agreements: Stockholm Conference on Human Environment 1972, Montreal Protocol, 1987, Conference of Parties (COPs, Kyoto Protocol, 1997, Clean Development Mechanism (CDM)

UNIT	TITLE	PERIODS
II	**Natural resources and its conservation**	6

Natural resources – types - Forest resources - Uses and over exploitation - Water Resources – Uses and over utilization - Water conservation- Rain Water Harvesting- Watershed Management – Mineral resources –Uses and Exploitation (Al, Cu, Fe), Food resources- World food problems - Effects of modern agriculture - Role of an individual in conservation of natural resources.

UNIT	TITLE	PERIODS
III	**Environmental Pollution and Global Issues**	6

Environmental pollution- types, causes, effects and controls; Air, water, soil - classification of pollutants based on solubility - hydrophilic, hydrophobicand lipophilic pollutants; Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture.
Pollution Tragedies: Love canal, Bhopal Gas, Minamata and Flint water and chernobyl – Disaster management: cyclone and earthquake.

UNIT	TITLE	PERIODS
IV	**GEOSPATIAL TECHNOLOGY**	6

Remote Sensing: Introduction, definition, fundamentals, Sensors, Active and passive remote sensing; data products, remote sensing satellites, Indian remote sensing satellite system (IRS).
GIS: Introduction, components, functions, types of GIS, Data structures in GIS-Raster and Vector data analysis-advantages and disadvantages.

GPS: Introduction, functions and segments of GPS, factors affecting GPS data.

UNIT	TITLE	PERIODS
V	**GEOSPATIAL TECHNOLOGY IN ENVIRONMENTAL STUDIES**	6

Application of Geospatial Technology: Watershed studies, Flood studies, Health Issues, Utility studies, Security and Defense studies, Urban and infrastructure development studies, Disaster management.

TOTAL PERIODS:	30

TEXT BOOKS:	
1.	Benny Joseph, 'Environmental Science and Engineering', Tata McGraw-Hill, New Delhi, 2006.
2.	Ravikrishnan. A, Environmental Science & Engineering, Sri Krishna Publications, (New Edition).
3.	Joseph G. Fundamentals of remote sensing. Universities Press, Hyderabad.2003.

REFERENCE BOOKS:	
1.	Elliott, D. 1997. Sustainable Technology. Energy, Society and Environment (Chapter 3). New York, Routledge Press.
2.	R.K. Trivedi, 'Handbook of Environmental Laws, Rules, Guidelines,Compliances and Standards', Vol. I and II, Enviro Media.
3.	Rajagopalan, R, 'Environmental Studies-From Crisis to Cure', Oxford University Press 2005.
4.	Lillisand, T. M. and Keifer, R. W. (2007). Remote sensing and image interpretation. John Willey and Sons, USA.
5.	Barrett, E. C. and Curtis, L. F. (1999). Introduction to environmental remote sensing. Chapman and Hall Publishers, USA.

WEBSITES:	
1.	https://www.sciencedirect.com/journal/journal-of-environmental-management
2.	https://journals.e-palli.com/home/index.php/ajgt

JOURNALS:	
1.	Journal of Environmental Management
2.	• Environmental Science & Policy.
3.	American Journal of Geospatial Technology

COURSE DESIGNERS			
1.	Dr. C. Ravichandran	Professor and Head	Chemistry
2.	Dr. R. Anitha devi	Assistant Professor	Chemistry
3.	Dr. D. Sivasankaran	Assistant Professor	Chemistry
4.	Dr. K. Saravanan	Associate Professor	Chemistry

Recommended by Board of Studies	Date: 12.10.2023	Syllabus version	1.0
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231CSC201T	PROGRAMMING IN C (Common to B.E Computer Science Engineering, B.Tech Information Technology and B.Tech Artificial Intelligence and Data Science)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	Learn to think logically and write pseudo code or draw flow charts for problems.
2.	Be exposed to the syntax of C.
3.	To develop C Programs using basic programming constructs.
4.	Learn to use arrays and strings in C.
5.	To develop applications in C using functions , pointers and structures
6	To do input/output and file handling in C

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Execute simple programs using basic C programming concepts.	K3
CO2:	Apply arrays and strings for simple application development.	K3
CO3:	Solve complex problems using functions and pointers.	K3
CO4:	Organize heterogeneous data with structures and unions.	K4
CO5:	Select suitable file access techniques for data processing.	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	2	1	-	1	-	-	2	2	-	3
CO2	3	3	3	1	-	1	-	-	2	2	-	3
CO3	3	3	3	3	-	1	-	-	2	2	-	3
CO4	3	3	3	2	-	1	-	-	2	2	-	3
CO5	3	3	3	2	-	1	-	-	2	2	-	3

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	**C PROGRAMMING BASICS**	9
Introduction- Algorithm – Flow Charts – Pseudo Code - Structure of a C program – compiling and linking processes – Character set - Constants, Variables – Data Types – Expressions using operators in C – Managing Input and Output operations – Decision Making and Branching – Looping statements.		
UNIT	**TITLE**	**PERIODS**
II	**ARRAYS AND STRINGS**	9
Arrays: Initialization – Declaration – Accessing the array elements – Operations on arrays- One dimensional array - two dimensional arrays – Strings: String operations – String Arrays - Simple programs: sorting- searching – matrix operations.		
UNIT	**TITLE**	**PERIODS**
III	**FUNCTIONS AND POINTERS**	9
Functions: Introduction - Function prototype - function definition - function call and return statements – Recursive functions. Parameter passing: Pass by value - Pass by reference. Pointers: Pointer operators – Declaring the pointer variable - Pointer arithmetic Null pointer- Arrays and pointers – Array of pointers.		
UNIT	**TITLE**	**PERIODS**
IV	**STRUCTURES AND UNIONS**	9
Structures: Introduction - Need for structures –definition and declaration – Structure within structure – Structures and functions – Union: Definition and Declaration – Accessing the members of union - Programs using Structures and Unions – Scope of variables - Storage classes - Preprocessor directives.		
UNIT	**TITLE**	**PERIODS**
V	**FILE HANDLING**	9
Introduction – Using files in C - File operation: Read data from files, writing data to files, detecting the end of file, Functions for selecting a record randomly – File pointer – Error handling - Types of file processing: Sequential access, Random access- Dynamic memory allocation.		
TOTAL PERIODS:		**45**

TEXT BOOKS:	
1.	Balagurusamy E — Programming in ANSI C, McGraw Hill Publication, Eighth Edition, 2019.
2.	Herbert Schildt — The Complete Reference C, McGraw Hill Education, Fourth Edition , 2017
3.	Reema Thareja, —Programming in C, Oxford University Press, Second Edition, 2016.

REFERENCE BOOKS:	
1.	Byron S. Gottfried, "Schaum's Outline of Theory and Problems of Programming with C", McGraw-Hill Education, 1996.
2.	Pradip Dey, Manas Ghosh, —Fundamentals of Computing and Programming in C, First Edition, Oxford University Press, 2009.
3.	Paul Deitel and Harvey Deitel, —C How to Program, Seventh edition, Pearson Publication
4.	Juneja, B. L and Anita Seth, —Programming in C, CENGAGE Learning India pvt. Ltd., 2011.
5.	Kernighan, B.W and Ritchie, D.M, —The C Programming language, Second Edition, Pearson Education,

	2006.

WEBSITES:	
1.	www.w3schools.com/c
2.	https://www.programiz.com/c-programming

JOURNALS:	
1.	Nil

EXTENSIVE READER:	
1.	Nil

COURSE DESIGNERS			
1.	Dr G S Anandha Mala	Professor	Computer Science and Engineering
2.	Dr S Ahamed Ali	Assistant Professor	Computer Science and Engineering
3.	Dr A Arockia Abins	Assistant Professor	Computer Science and Engineering
4.	Dr J Deepa	Associate Professor	Computer Science and Engineering
5	Dr V Balaji	Associate Professor	CSE (Cyber Security)

Recommended by Board of Studies	Date: 20-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231GEH201T	Tamils and Technology/தமிழரும்தொழில்நுட்பமும் (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		1	0	0	0	1

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	40 %	60 %		-	45%	50%

PREREQUISITES:

Nil

COURSE OBJECTIVES:

1.	தமிழர்கள்நெசவுமற்றும்பானைத்தொழிற்நுட்பங்களில்சிறந்துவிளங்கியிருப்பதைஅறிய வைத்தல்/To know that Tamils have excelled in weaving and pottery techniques.
2.	கட்டிட வடிவமைப்பு மற்றும் கட்டிடத் தொழிற்நுட்பங்களில் தமிழர்கள் சிறந்து விளங்கியதை எடுத்துக்காட்டுகளுடன் அறிய வைத்தல்./ To Exemplify Tamil excellence in architectural design and construction techniques.
3.	கப்பல் கட்டுதல், இரும்புத் தொழில், நாணயம் அச்சிடுதல், மணி தயாரித்தல் போன்ற உற்பத்தித் தொழிற்நுட்பங்களில் தமிழர்கள் திறனுற்று விளங்கியதை அறிய வைத்தல்/ To show that Tamils were skilled in manufacturing techniques such as ship building, iron industry, coin minting, bell making etc.
4.	அணை, எரி, குளம், கிணறு போன்றவற்றை வடிவமைத்தல் மூலமாகவும் பராமரித்தல் மூலமாகவும் தமிழர்கள் வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழிற்நுட்பத்தில் சிறந்து விளங்கியதை அறிய வைத்தல்/ To promote Tamil excellence in agriculture and irrigation technology through design and maintenance of dams, streams, ponds, wells etc.
5.	அறிவியல் தமிழ் மற்றும் கணித் தமிழ் வளர்ச்சியையும் தமிழ் மென்பொருள் உருவாக்கம் பற்றியும் அறிய வைத்தல்/ To know about Scientific Tamil and Digital Tamil development and Tamil software development.

COURSE OUTCOMES (COs):

இந்தப்பாடத்தினைகற்றுமுடிக்கும்போது,மாணவர்கள்கீழ்க்கண்டவற்றைபற்றிஅறிந்திருப்பார்கள்:.Upon completion of this course, students will be aware of the following:		Bloom's level
CO1:	தமிழர்கள்நெசவுமற்றும்பானைத்தொழிற்நுட்பங்களில்சிறந்துவிளங்கியிருப்பதைப் பற்றி./ About Tamils excelling in weaving and pottery techniques.	K2
CO2:	கட்டிட வடிவமைப்பு மற்றும் கட்டிடத் தொழிற்நுட்பங்களில் தமிழர்கள் சிறந்து விளங்கியது பற்றி./ About the excellence of Tamils in architectural design and construction techniques.	K2
CO3:	கப்பல் கட்டுதல், இரும்புத் தொழில், நாணயம் அச்சிடுதல், மணி தயாரித்தல் போன்ற உற்பத்தித் தொழிற்நுட்பங்களில் தமிழர்கள் திறனுற்று விளங்கியதைப் பற்றி./ About Tamils becoming proficient in manufacturing techniques like ship building, iron industry, coinage, bell making etc.	K2

CO4:	அணை, எரி, குளம், கிணறு போன்றவற்றை வடிவமைத்தல் மூலமாகவும் பராமரித்தல் மூலமாகவும் தமிழர்கள் வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழிற்நுட்பத்தில் சிறந்து விளங்கியதைப் பற்றி./About how Tamils excelled in agriculture and irrigation technology by designing and maintaining dams, streams, ponds, wells etc.	K2
CO5:	அறிவியல் தமிழ் மற்றும் கணித் தமிழ் வளர்ச்சி பற்றியும் தமிழ் மென்பொருள் உருவாக்கம் பற்றியும் / About Development of Scientific Tamil and Digital Tamil and Tamil Software Development	K2

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	**WEAVING AND CERAMIC TECHNOLOGY /நெசவுமற்றும்பானைத்தொழில்நுட்பம்**	3

Weaving Industry during Sangam Age – Ceramic technology – Black and Red Ware Potteries (BRW) – Graffiti on Potteries

சங்ககாலத்தில்நெசவுத்தொழில் - பானைத்தொழில்நுட்பம் - கருப்புசிவப்புபாண்டங்கள் - பாண்டங்களில்கீறல்குறியீடுகள்.

UNIT	TITLE	PERIODS
II	**DESIGN AND CONSTRUCTION TECHNOLOGY / வடிவமைப்புமற்றும்கட்டிடத்தொழில்நுட்பம்**	3

Designing and Structural construction House & Designs in household materials during Sangam Age - Building materials and Hero stones of Sangam age – Details of Stage Constructions in Silappathikaram - Sculptures and Temples of Mamallapuram - Great Temples of Cholas and other worship places - Temples of Nayaka Period - Type study (Madurai Meenakshi Temple)- Thirumalai Nayakar Mahal - Chetti Nadu Houses, Indo - Saracenic architecture at Madras during British Period.

சங்ககாலத்தில்வடிவமைப்புமற்றும்கட்டுமானங்கள்&சங்ககாலத்தில்வீட்டுப்பொருட்களில்வடிவமைப்பு - சங்ககாலத்தில்கட்டுமானபொருட்களும்நடுகல்லும் - சிலப்பதிகாரத்தில்மேடைஅமைப்புபற்றியவிவரங்கள் - மாமல்லபுரச்சிற்பங்களும்கோவில்களும் - சோழர்காலத்துப்பெருங்கோயில்கள்மற்றும்பிறவழிபாட்டுத்தலங்கள் - நாயக்கர்காலக்கோயில்கள் - மாதிரிகட்டமைப்புகள்பற்றிஅறிதல்,
மதுரைமீனாட்சிஅம்மன்ஆலயம்மற்றும்திருமலைநாயக்கர்மஹால் – செட்டிநாட்டுவீடுகள் - பிரிட்டிஷ்காலத்தில்சென்னையில்இந்தோ-சாரோசெனிக்கட்டிடக்கலை.

UNIT	TITLE	PERIODS
III	**MANUFACTURING TECHNOLOGY / உற்பத்தித்தொழில்நுட்பம்**	3

Art of Ship Building - Metallurgical studies - Iron industry - Iron smelting, steel -Copper and goldCoins as source of history - Minting of Coins – Beads making-industries Stone beads -Glass beads - Terracotta beads -Shell beads/ bone beats - Archeological evidences - Gem stone types described in Silappathikaram.

கப்பல்கட்டும்கலை – உலோகவியல் - இரும்புத்தொழிற்சாலை - இரும்பைஉருக்குதல், எஃகு - வரலாற்றுச்சான்றுகளாகசெம்புமற்றும்தங்கநாணயங்கள் - நாணயங்கள்அச்சடித்தல் - மணிஉருவாக்கும்தொழிற்சாலைகள் - கல்மணிகள், கண்ணாடிமணிகள் - சுடுமண்மணிகள் - சங்குமணிகள் - எலும்புத்துண்டுகள் - தொல்லியல்சான்றுகள் - சிலப்பதிகாரத்தில்மணிகளின்வகைகள்.

UNIT	TITLE	PERIODS
IV	**AGRICULTURE AND IRRIGATION TECHNOLOGY /வேளாண்மைமற்றும்நீர்ப்பாசனத்தொழில்நுட்பம்**	3

Dam, Tank, ponds, Sluice, Significance of Kumizhi Thoompu of Chola Period, Animal Husbandry - Wells designed for cattle use - Agriculture and Agro Processing - Knowledge of Sea - Fisheries – Pearl - Conche diving - Ancient Knowledge of Ocean - Knowledge Specific Society.

அணை, எரி, குளங்கள், மதகு - சோழர்காலக்குமுத்நூம்பின்முக்கியத்துவம் - கால்நடைபராமரிப்பு - கால்நடைகளுக்காகவடிவமைக்கப்பட்டகிணறுகள் - வேளாண்மைமற்றும்வேளாண்மைச்சார்ந்தசெயல்பாடுகள் - கடல்சார்அறிவு - மீன்வளம் - முத்துமற்றும்முத்துக்குளித்தல் - பெருங்கடல்குறித்தபண்டையஅறிவு - அறிவுசார்சமூகம்.

UNIT	TITLE	PERIODS
V	**SCIENTIFIC TAMIL & TAMIL COMPUTING /அறிவியல்தமிழ்மற்றும்கணித்தமிழ்**	3

Development of Scientific Tamil - Tamil computing – Digitalization of Tamil Books – Development of Tamil Software – Tamil Virtual Academy – Tamil Digital Library – Online Tamil Dictionaries – Sorkuvai Project.

அறிவியல்தமிழின்வளர்ச்சி - கணித்தமிழ்வளர்ச்சி - தமிழ்நூல்களைமின்பதிப்புசெய்தல் - தமிழ்மென்பொருட்கள்உருவாக்கம் - தமிழ்இணையக்கல்விக்கழகம் - தமிழ்மின்நூலகம்இணையத்தில்தமிழ்அகராதிகள்- சொற்குவைத்திட்டம்.

TOTAL PERIODS:	**15**

TEXT BOOKS CUM REFERENCE BOOKS:

1.	தமிழகவாலாறுமக்களும்பண்பாடும்கே.கே. பிள்ளை (வெளியீடு: தமிழ்நாடுபாடநூல்மற்றும்கல்வியியல்பணிகள்கழகம்).
2.	கணினித்தமிழ் - முனைவர்இல. சுந்தரம். (விகடன்பிரசுரம்).
3.	வைகைநதிக்கரையில்சங்ககாலநகரநாகரிகம் (தொல்லியல்துறைவெளியீடு)
4.	பொருறை - ஆற்றங்கரைநாகரிகம், தொல்லியல்துறைவெளியீடு)
5.	Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6.	Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
7.	Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8.	The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
9.	Keeladi - 'Sangam City Civilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10.	Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by: The Author)
11.	Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12.	Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

COURSE DESIGNERS

1.	Anna University, Chennai

Recommended by Board of Studies	Date:	Syllabus version	-
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

231PYB011L	**PHYSICS LABORATORY** (Common to all branches of Engineering and Technology)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		**0**	**0**	2	0	1

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	60 %	40 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To impart knowledge on experimental skills to determineelastic, optical and thermal properties of materials.
2.	To demonstrate the experimental determination of compressibility of liquid and band gap of a semiconductor.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	determine the elastic properties of materials using torsional pendulum and non uniform bending.	K3
CO2:	determine the optical properties of light waves using optical fibre, laser and spectrometer.	K3
CO3:	determine the physical properties of materials using Lee's disc apparatus and air wedge.	K3
CO4:	calculate the compressibility of liquid using ultrasonic interferometer.	K3
CO5:	calculate the band gap of a semiconductor using band gap apparatus.	K3

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**

CO1	3	2	1	-	-	-	-	-	-	-	-	1
CO2	3	2	1	-	-	-	-	-	-	-	-	1
CO3	3	2	1	-	-	-	-	-	-	-	-	1
CO4	3	2	1	-	-	-	-	-	-	-	-	1
CO5	3	2	1	-	-	-	-	-	-	-	-	1
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

ANY FIVE EXPERIMENTS	
S. No	**NAME OF THE EXPERIMENT**
1	Torsion Pendulum – Rigidity modulus of wire and moment of inertia of disc
2	Non Uniform Bending – Determination of Young's modulus
3	Semiconductor Laser –Wavelength of laser light and Size of particle
4	Optical Fiber – Numerical Aperture and Acceptance Angle
5	Lee's Disc method - Determination of thermal conductivity of a bad conductor
6	Spectrometer – Dispersive power of the prism
7	Air Wedge – Measurement of thickness of thin wire
EXPERIMENTS BEYOND THE SYLLABUS	
ANY ONE EXPERIMENT	
1	Ultrasonic Interferometer - Velocity of sound and Compressibility of liquid
2	Determination of the Band gap of a semiconductor

TOTAL PERIODS:	**15**

REFERENCE BOOKS:	
1	Physics Laboratory Manual, Department of Physics, Easwari Engineering College.
2	R.K.Shukla and Anchal Srivastava, Practical Physics, 1st Edition, New Age International (P) Ltd, New Delhi, 2006.
3	G.L.Souires, Practical Physics, 4th Edition, Cambridge University, UK, 2001.
4	D.Chattopadhyay, P.C. Rakshit and B.Saha, An Advanced Course in Practical Physics, 2nd ed., Books & Allied Ltd., Calcutta, 1990.

COURSE DESIGNERS			
1.	Dr. S. Nirmala	Assoc Professor & Head	Department of Physics
2.	Dr. G. Rajkumar	Professor	Department of Physics
3.	Dr. R. Sivakumar	Asst Professor	Department of Physics
4.	Dr. K. Raju	Asst Professor	Department of Physics

Recommended by Board of Studies	Date: 12-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6



231GES012L	VISUALISATION OF DESIGN AND DRAWING (Common to AI&DS, CSE, CSE (AIML), CSE (CS), CSBS, CSD, ECE, EEE, IT	Periods per week				Credits
		L	T	P	R	
Regulation - R23		0	0	3	1	2

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous assessment Examination	End Semester Examination		Continuous assessment Examination	End Semester Examination	
3	60 %	40 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To develop students, graphic skills for communication of concepts, ideas and design of engineering products
2.	To expose them to existing National standards related to technical drawings.
3.	To familiarize with basic geometrical constructions and orthographic projections using drafting software.
4.	To make the students to draw the different projections of the solids.
5.	To get an idea about 3D views through isometric projections.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Apply BIS and ISO standards in Engineering drawing.	Apply
CO2:	Use the CAD software for drawing the plane surfaces by rotating plane method.	Apply
CO3:	Apply the Projection concepts and drafting tools to the projection of simple solids.	Apply
CO4:	Construct orthographic projections from pictorial views of simple machine elements.	Apply
CO5:	Draw the pictorial views of prism, pyramid, cylinder, cone and combination of solids.	Apply

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	3	1	-	1	2	2	-	1
CO2	3	3	2	-	3	1	-	1	2	2	-	1
CO3	3	3	2	-	3	1	-	1	2	2	-	1
CO4	3	3	2	-	3	1	-	1	2	2	-	1

CO5	3	3	2	-	3	1	-	1	2	2	-	1

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	BASIC CONVENTIONS USED IN ENGINEERING DRAWING AND INTRODUCTION TO COMPUTER GRAPHICS	12

Principles of Engineering graphics and their significance - Use of drawing Instruments-BIS conventions and Specifications-Size, Layout and folding of drawing Sheets-Lettering and Dimensioning.

The Concepts of Computer Aided Drafting for Engineering drawing, Computer graphics & Geometrical modeling (2D Orthographic Views) and 3D drafting (Isometric Views) using AutoCAD.

List of Experiments:

1. Drawing of a title block with necessary text, projection symbol and lettering using CAD software.
2. Drawing of line, Planes (Polygonal and Circular surfaces).

UNIT	TITLE	PERIODS
II	PROJECTION OF PLANES	12

Projection of planes (polygonal and circular surfaces) inclined to one of the principal planes by rotating object method.

List of Experiments:

1. Drawing of plane Surface inclined to HP.
2. Drawing of plane Surface inclined to VP.

UNIT	TITLE	PERIODS
III	PROJECTION OF SOLIDS	12

Projection of simple solids like Prisms, Pyramids, Cylinder and Cone when the axis is inclined to one of the principal planes by rotating object method.

List of Experiments:

1. Drawing of simple solids like prism and pyramids when the axis is inclined to HP or VP.
2. Drawing of simple solids like cylinder and cone when the axis is inclined to HP or VP.

UNIT	TITLE	PERIODS
IV	ORTHOGRAPIC PROJECTION	12

Visualization concepts and Orthographic Projection - Layout of views – Orthographic Projection- Conversion of pictorial diagram into orthographic views

List of Experiments:

1. Drawing orthographic view of simple solids like Prism, Pyramids, Cylinder, Cone, etc, and dimensioning.
2. Drawing of orthographic views from the given pictorial diagram.

UNIT	TITLE	PERIODS
V	ISOMETRIC DRAWING	12

Principles of isometric view – Isometric view of simple solids – Prism, Pyramid, Cylinder, Cone, and combination of solids.

List of Experiments:

1. Drawing isometric projection of simple solids.

2.	Modeling of 2D to 3D objects using drafting software

TOTAL PERIODS:	**60**

TEXT BOOKS:	
1.	Natarajan K.V., "A text book of Engineering Graphics", Dhanalakshmi Publishers, Chennai, 2009
2.	Jayapoovan T, "Engineering Graphics using AUTOCAD", Vikas Publishing ,7 th Edition.
3.	Venugopal K. and Prabhu Raja V., "Engineering Drawing with AUTOCAD and building drawing", New Age International (P) Limited, 2018, 5th edition.

REFERENCE BOOKS:	
1.	Basant Agarwal and Agarwal C.M., "Engineering Drawing", Tata McGraw Hill Publishing Company Limited, New Delhi, 2008.
2.	Bhatt N.D. and Panchal V.M., "Engineering Drawing", Charotar Publishing House, 50th Edition, 2010.
3.	Dinesh Kumar S, K.Sivakumar and R.Ramadoss, " Engineering Graphics", Maruthi Publishers, Chennai,2019.
4.	Gopalakrishna K.R., "Engineering Drawing" (Vol. I&II combined), Subhas Stores, Bangalore, 2007.
5.	Parthasarathy N S and Vela Murali, "Engineering Graphics", Oxford University, Press, New Delhi, 2015.
6.	Shah M.B., and Rana B.C., "Engineering Drawing", Pearson, 2nd Edition, 2009.
7.	CAD software Theory and User Manuals.

WEBSITES:	
1.	https://ieeexplore.ieee.org/abstract/document/6132125
2.	https://www.indianjournals.com/ijor.aspx?target=ijor:ajmr&volume=10&issue=1&article=039

JOURNALS (Publication of Bureau of Indian Standards):	
1.	IS 10711 – 2001: Technical products Documentation – Size and lay out of drawing sheets
2.	IS 9609 (Parts 0 & 1) – 2001: Technical products Documentation – Lettering.
3.	IS 10714 (Part 20) – 2001 & SP 46 – 2003: Lines for technical drawings
4.	IS 11669 – 1986 & SP 46 – 2003: Dimensioning of Technical Drawings.
5.	IS 15021 (Parts 1 to 4) – 2001: Technical drawings – Projection Methods.

COURSE DESIGNERS			
1.	Dr. R. Ramadoss	Professor	Mechanical Engineering
2.	Mr. R. Jeremiah	Assistant Professor	Mechanical Engineering
3.	Mr. M. Raju	Assistant Professor	Mechanical Engineering

Recommended by Board of Studies	Date: 16-11-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6



231CSC211L	**C PROGRAMMING LABORATORY** (Common to CSE, IT, CSE (AIML), CSE (CS) and AI-DS)	Periods per week				Credits
		L	T	P	R	
Regulation - R23		**0**	**0**	**4**	**0**	**2**

SCHEME OF EXAMINATION						
Duration of End Semester Examination in Hours	Maximum Marks - 100			Minimum Marks for Pass - 50		
	Weightage			Weightage		Overall
	Continuous Assessment Examination	End Semester Examination		Continuous Assessment Examination	End Semester Examination	
3	60 %	40 %		-	45%	50%

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To develop programs in C using basic constructs.
2.	To develop applications in C using strings and pointers.
3.	To develop applications in C using functions and structures.
4.	To develop applications in C using file processing.
5.	To develop simple project using C

COURSE OUTCOMES:		Bloom's level
Upon completion of this course, student will be able to:		
CO1:	Develop simple programs using basic constructs in C programming.	K5
CO2:	Develop C programs for simple applications making use of arrays and strings.	K5
CO3:	Implement modular programming with functions.	K3
CO4:	Build programs with storage classes and pointers for memory management.	K6
CO5:	Construct programs with user defined data types.	K6
CO6:	Design applications using file processing techniques.	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	2	3	1	1	1	2	2	3	2
CO2	3	3	3	3	3	2	2	2	2	2	3	2
CO3	3	3	3	3	3	2	2	2	3	3	3	2
CO4	3	3	3	3	3	2	2	2	3	3	3	2
CO5	3	3	3	2	3	1	1	2	2	3	3	2
CO6	3	3	3	2	3	1	1	2	2	3	3	2
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

LIST OF EXPERIMENTS	
1.	Programs using only I/O functions.
2.	Programs to study operators and data types.
3.	Programs based on control structures (IF, SWITCH CASE).
4.	Programs using FOR and WHILE loops.
5.	Programs using single dimensional arrays.
6.	Programs using multi-dimensional arrays.
7.	Programs on Sorting and Searching using arrays.
8.	Programs based on String manipulations.
9.	Programs based on User Defined Functions.
10.	Programs using Functions with Parameters.
11.	Programs using Storage Classes.
12.	Programs to introduce Pointers.
13.	Programs using Structures and Union.
14.	Programs using Array of Structures.
15.	Programs based on Files.
16.	Mini Project
TOTAL PERIODS:	**60**

CONTENT BEYOND SYLLABUS	
1.	Programs using basic graphic functions
2.	Programs using Conditional Compilation directives

LIST OF EQUIPMENTS:	
1.	**HARDWARE**:
• Standalone desktops. 30nos.	
2.	**SOFTWARE**
• GCC Compiler , Turbo C or equivalent	

COURSE DESIGNERS			
1.	Dr G S Anandha Mala	Professor	Computer Science and Engineering
2.	Dr S Ahamed Ali	Assistant Professor	Computer Science and Engineering
3.	Dr A Arockia Abins	Assistant Professor	Computer Science and Engineering
4	Dr J Deepa	Associate Professor	Computer Science and Engineering
5.	Dr V Balaji	Associate Professor	CSE (Cyber Security)

Recommended by Board of Studies	Date: 20-10-2023	Syllabus version	1
Approved by the Academic Council	Date: 24-01-2024	Meeting No.	6

CURRICULUM AND SYLLABUS

(REGULATIONS 2023 V1.0)

FOR

B.E. COMPUTER SCIENCE AND ENGINEERING

CHOICE BASED CREDIT SYSTEM

(Applicable to the students admitted from the Academic Year 2023-24 onwards)



EASWARI ENGINEERING COLLEGE

(Autonomous Institution)
Bharathi Salai, Ramapuram, Chennai - 600 089

R2023 V1.0

SEMESTER III									
S.No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231MAB302T	Discrete Mathematics	BS	3	1	-	-	4	40 / 60
2	231CSS301T	Digital Principles and Computer Organization	ES	3	-	-	-	3	40 / 60
3	231CSC301T	Data Structures and Algorithms	PC	3	-	-	-	3	40 / 60
4	231CSC302T	Object Oriented Programming with C++	PC	3	-	-	-	3	40 / 60
5	231CSC303T	Foundations of Data Science	PC	3	-	-	-	3	40 / 60
LABORATORY									
7	231CSC311L	Data Structures Laboratory	PC	-	-	4	0	2	60 / 40
8	231CSC312L	Object Oriented Programming Laboratory	PC	-	-	3	1	2	60 / 40
9	231CSC313L	Data Science Laboratory	PC	-	-	4	0	2	60 / 40
10	231SDA001	Comprehensive Programming - Foundations	EEC	0	0	2	0	0	-
TOTAL CREDITS								**22**	**-**

SYLLABUS OF

SEMESTER – III

COURSES

231MAB302T	DISCRETE MATHEMATICS	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0	(Common to AI & DS, CSE (AI & ML), CSE (CS), CSE, IT & RA)	**3**	**1**	**0**	**0**	**4**

SCHEME OF EXAMINATION

Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

COURSE OBJECTIVES:

1.	To extend student's logical and mathematical maturity and ability to deal with abstraction.
2.	To introduce most of the basic terminologies used in computer science courses and application of ideas to solve practical problems.
3.	To understand the basic concepts of Combinatorics and graph theory.
4.	To familiarize the applications of algebraic structures.
5.	To understand the concepts and significance of lattices and Boolean algebra which are widely used in computer science and engineering.

COURSE OUTCOMES (COs):

Upon completion of this course, student will be able to:		Bloom's Level
CO1:	The verify validity of the given propositions by inference theory.	K5
CO2:	Use basic counting principles to get the solution for problems encountered in real life situation.	K4
CO3:	Apply Graph theoretical ideas to find the number of paths between two nodes.	K4
CO4:	Derive group theoretic results and properties.	K6
CO5:	Find the least upper bound and the greatest lower bound of a given Lattice and also simplify Boolean expressions.	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	2
CO2	3	3	3	-	-	-	-	-	-	-	-	2
CO3	3	3	3	-	-	-	-	-	-	-	-	2
CO4	3	3	3	-	-	-	-	-	-	-	-	2
CO5	3	3	3	-	-	-	-	-	-	-	-	2

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	MATHEMATICAL LOGIC	PERIODS
I	Statements and Notations – Connectives – Normal forms – Theory of inference for the statement calculus – Predicate calculus – Inference theory of the predicate calculus	12
UNIT	COMBINATORICS	PERIODS
II	Mathematical induction – Strong induction – The basics of counting – The pigeonhole principle – Permutations and combinations – Recurrence relations – Solving linear recurrence relations – Generating functions – Inclusion and exclusion principle and its applications.	12
UNIT	GRAPHS	PERIODS
III	Graphs and graph models – Graph terminology and special types of graphs – Matrix representation of graphs and graph isomorphism – Connectivity – Euler and Hamilton paths.	12
UNIT	ALGEBRAIC STRUCTURES	PERIODS
IV	Groups – Subgroups – Homomorphism ‘s – Normal subgroup and cosets – Lagrange ‘s theorem – Definitions and examples of Rings, Integral domains and Fields.	12
UNIT	LATTICES AND BOOLEAN ALGEBRA	PERIODS
V	Partial ordering – Posets – Lattices as posets – Properties of lattices - Some special lattices – Boolean algebra.	12
TOTAL PERIODS:		60

TEXT BOOKS:	
1.	Rosen, K.H., "Discrete Mathematics and its Applications", 7th Edition, Tata McGraw Hill Pub. Co. Ltd., New Delhi, Special Indian Edition, 2011.
2.	Tremblay, J.P. and Manohar.R, “Discrete Mathematical Structures with Applications to Computer Science", Tata McGraw Hill Pub. Co. Ltd, New Delhi, 30th Reprint,
3.	Work-Book on“ Discrete Mathematics”, Chess Educational Publishers, prepared by Department of Mathematics

REFERENCE BOOKS:	
1.	Grimaldi, R.P. "Discrete and Combinatorial Mathematics: An Applied Introduction", 4th Edition, Pearson Education Asia, Delhi, 2007.
2.	Lipschutz, S. and Mark Lipson., "Discrete Mathematics", Schaum‘s Outlines, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 3rd Edition, 2010.
3.	Koshy, T. "Discrete Mathematics with Applications", Elsevier Publications, 2006.

WEBSITES:	
1.	http://en.m.wikipedia.org
2.	http://www.coursera.org
3.	http://discrete.openmathbooks.org

JOURNALS:	
1.	Nil

EXTENSIVE READER:	
1.	Discrete Mathematical Structures, Kolman, R.C.Busby and S.C.Ross, 6th Edition, PHI, 2018
2.	Discrete Mathematics, Richard Johnsonbaugh, 8th Edition, Prentice Hall, 2017.
3.	Elements of Discrete Mathematics–A Computer Oriented Approach, C.L.Liu, Tata McGraw Hill, Special Indian Edition, 2017

COURSE DESIGNERS			
1.	Dr.S.Muthukumar	Professor	Department of Mathematics
2.	Dr.S.R.Ananthalakshmi	Associate Professor & HOD	Department of Mathematics
3.	Dr. C. Subramanian	Asst. Prof. (Sr. Gr.)	Department of Mathematics

Recommended by Board of Studies	Date: 12.10.2023	Syllabus version	V.1
Approved by the Academic Council	Date: 24.01.2024	Meeting No.	6

231CSS301T	DIGITAL PRINCIPLES AND COMPUTER ORGANIZATION (Common to CSE (AI & ML), CSE (CS), CSE, IT)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	Develop a comprehensive understanding of digital principles.
2.	Analyze and apply Boolean algebra to design and optimize digital circuits.
3.	Explore the architecture and components of modern computer systems.
4.	Investigate various memory technologies and their organization within computer systems.
5.	Examine the organization and execution of instructions within a CPU, including instruction set architecture (ISA) and execution models.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Understand the foundational principles of digital systems and their applications.	K2
CO2:	Design and analyze digital circuits using Boolean algebra and logic gates.	K6
CO3:	Describe the architecture and functionalities of key components in computer systems, including CPUs.	K1
CO4:	Explain the organization and operation of different types of memory in computer systems.	K2
CO5:	Evaluate the performance implications of instruction set designs.	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	2	-	2	-	-	-	-	-	-	2
CO2	3	3	2	-	2	-	-	-	-	-	-	2
CO3	3	1	2	-	1	-	-	-	-	-	-	2
CO4	3	1	2	-	1	-	-	-	-	-	-	2
CO5	3	2	2	-	2	-	-	-	-	-	-	2
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

UNIT	Introduction to Digital Principles	PERIODS
I	Number System - Number Base Conversion - Number Representation - Binary Addition - Binary Subtraction - Floating Point Representation - BCD - Codes : ASCII Unicode - Parity Bit - Digital Logic - Basic Gates - Universal Logic Gates	9
UNIT	Combinational Logic and Data Processing Circuit	PERIODS
II	Boolean Algebra Theorems - Minterms - Maxterms - Karnaugh Map (K-Map) - Sum of Products (SOP) and Product of Sums (POS) - Decoder - Encoder - Multiplexer (MUX) - Demultiplexer (DMUX) - Seven-Segment Display	10
UNIT	Introduction to Computer Architecture	PERIODS
III	Components of a Microcomputer - Central Processing Unit (CPU) - CPU technology - CPU Architecture - Intel Microprocessor Family - Multicore Processors - CPU Instruction Execution - Disk Controller - Microcomputer Bus - USB Architecture - PCI Express Bus - FireWire	9
UNIT	Memory	PERIODS
IV	Introduction - Semiconductor Memory - Hard Disk - Solid-State Drive (SSD) - Memory Hierarchy - Cache Memory - Mapping Methods - Virtual Memory - Memory Organization of a Computer	8
UNIT	Assembly Language and ARM Instructions	PERIODS
V	Introduction - Instruction Set Architecture (ISA) - ARM Processor Architecture - ARM Registers - ARM Instructions - ARM Data Processing Instruction Format - ARM Data Transfer Instructions - ARM Addressing Mode	9
	TOTAL PERIODS	45

TEXT BOOKS:	
1.	Computer Systems: Digital Design, Fundamental of Computer Architecture and Assembly Language, Ata Elachi, Springer, 2018. ISBN 978-3-319-66774-4

REFERENCE BOOKS:	
1.	"Computer Systems: A Programmer's Perspective" by Randal E. Bryant and David R. O'Hallaron, Pearson Education, Third edition, 2016.
2.	"Fundamentals of Digital Logic with Verilog Design" by Stephen Brown and ZvonkoVranesic, McGraw-Hill Education, Third edition, 2014.
3.	"Computer Organization and Design: The Hardware/Software Interface" by David A. Patterson and John L. Hennessy, Morgan Kaufmann Publishers, Fifth edition, 2013.
4.	"Digital Design and Computer Architecture" by David Money Harris and Sarah L. Harris, Morgan Kaufmann Publishers, Second edition, 2012.
5.	"Structured Computer Organization" by Andrew S. Tanenbaum, Pearson Education, Sixth edition, 2012.

WEBSITES:	
1.	https://www.keil.com/
2.	https://www.gnu.org/

JOURNALS:	
1.	Nil

EXTENSIVE READER:	
1.	https://www.arm.com/architecture/cpu

COURSE DESIGNERS			
1.	Dr A ArockiaAbins	Assistant Professor	Computer Science and Engineering
2.	Dr S Ahamed Ali	Assistant Professor	Computer Science and Engineering
3.	Dr R Sheeja	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	1
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC301T	DATA STRUCTURES AND ALGORITHMS (Common to AI & DS, CSE (AI & ML), CSE (CS), CSE, IT)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To understand the concepts of ADTs
2.	To Learn linear data structures – lists, stacks, and queues
3.	To understand sorting, searching and hashing algorithms
4.	To apply Tree and Graph structures
5.	To Learn Algorithm Design Techniques

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Infer the concepts of abstract data types for linear data structures.	K4
CO2:	Apply non-linear data structures concepts in real time applications	K3
CO3:	Develop solutions using linear and non- linear data structures.	K3
CO4:	Illustrate various searching and sorting techniques to solve the complex problem.	K4
CO5:	Evaluate space and time complexity for a given algorithm	K6

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	1	2	2	1	-	-	-	-	-	2	1	1
CO2	1	2	1	1	-	-	-	-	-	1	-	1
CO3	2	2	1	1	-	-	-	-	-	1	-	1
CO4	2	2	1	1	-	-	-	-	-	1	-	1
CO5	1	3	2	2	-	2	1	-	-	1	1	1
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

UNIT	TITLE	PERIODS
I	**LINEAR DATA STRUCTURES – LIST**	9
Introduction to Data Structure- Abstract Data Types (ADTs) – List ADT – Array-based implementation – linked list implementation —singly linked lists- circularly linked lists- Doubly-linked lists – Stack ADT- Queue ADT- Applications of Stack.		
UNIT	**TITLE**	**PERIODS**
II	**NON-LINEAR DATA STRUCTURES – TREES**	9
Tree ADT –Tree terminologies- Binary Tree ADT-Tree traversals – Expression trees – Binary search tree ADT– Threaded Binary Trees- AVL Trees – B-Tree -applications of trees -Heap–Binary Heap- Applications of heap.		
UNIT	**TITLE**	**PERIODS**
III	**NON-LINEAR DATA STRUCTURES - GRAPHS**	9
Definition – Representation of Graph – Types of graph - Breadth-first traversal - Depth-first traversal – Topological Sort – Applications of graphs-minimum spanning tree-Dijkstra Algorithm-Kruskal Algorithm.		
UNIT	**TITLE**	**PERIODS**
IV	**SEARCHING AND SORTING**	9
Searching-Introduction- Linear Search - Binary Search. Sorting - Bubble sort –Quick Sort - Insertion sort - Merge sort-Shell sort-Selection sort.		
UNIT	**TITLE**	**PERIODS**
V	**HASHING AND ALGORITHM DESIGN TECHNIQUES**	9
Hashing-Hash Functions – Separate Chaining – Open Addressing – Rehashing – Double Hashing-Extendible Hashing- Dynamic programming – Knapsack problem -Recursive Algorithms.		

TOTAL PERIODS	**45**

TEXT BOOKS:	
1.	Mark Allen Weiss, “Data Structures and Algorithm Analysis in C”, 2nd Edition, Pearson Education, 2020.

REFERENCE BOOKS:	
1.	Thomas H. Cormen, Charles E. Leiserson, Ronald L.Rivest, Clifford Stein, “Introduction to Algorithms fourth Edition, Mcgraw Hill, 2022.
2.	Aho, Hopcroft and Ullman, “Data Structures and Algorithms”, Pearson Education, 2009.
3.	Stephen G. Kochan, “Programming in C”, 3rd edition, Pearson Education.
4.	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, “Introduction to Algorithms", Second Edition, McGraw Hill, 2002.

COURSE DESIGNERS			
1.	Mrs. HemaPriya K	Assistant Professor	Computer Science and Engineering
2.	Dr J Deepa	Associate Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	1
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC302T	OBJECT ORIENTED PROGRAMMING WITH C++	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEMEOFEXAMINATION

Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:

Programming in C

COURSE OBJECTIVES:

1.	To comprehend the fundamentals of object oriented programming in C++.
2.	To use object oriented programming to develop generic programming skills
3.	Apply appropriate OOP principles to solve complex problems
4.	Design programs that can handle run time errors and exceptions
5.	Develop programs using files and templates

COURSE OUTCOMES(COs):

Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Apply the concepts of object oriented programming	K3
CO2:	Apply OOP concepts to develop programs using functions and constructors in C++.	K3
CO3:	Incorporate the inheritance and polymorphism concepts to develop applications in C++	K5
CO4:	Apply operator overloading concepts in C++.	K3
CO5:	Exemplify the process of data file manipulations, templates and exception handling using C++	K5

MAPPING OF COURSE OUTCOMES(CO) WITH PROGRAMME OUTCOME(PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	-	-	-	2	-	2	3
CO2	3	3	3	2	2	-	-	-	2	-	2	3
CO3	3	3	3	2	2	-	-	-	2	-	2	3
CO4	3	3	3	2	2	-	-	-	2	-	2	3
CO5	3	3	3	3	1	2	2	1	2	2	2	3

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	INTRODUCTION	9
. Object oriented programming concepts: Class – Object - Abstraction – Encapsulation – Polymorphism- inheritance- Data Hiding, Introduction to C++:Structure of C++ program, Data types, Operators and control statements, Input and output operators, Dynamic initialization, Reference variables, Classes and Objects: Class specification- Member function definition		
UNIT	TITLE	PERIODS
II	FUNCTIONS, CONSTRUCTORS AND DESTRUCTORS	10
Function prototype, Default arguments, Call by reference-Static data members and Static member functions -Function return type as objects - Friend function – Constructors and destructors: Basic concepts - Parameterized constructor - Copy constructor - Dynamic constructors.		
UNIT	TITLE	PERIODS
III	INHERITANCE AND POLYMORPHISM	9
Defining derived classes, Types of inheritance: Single inheritance- Multilevel inheritance- Multiple inheritance – Hierarchical inheritance – Hybrid inheritance- Constructors in derived and base class, Abstract classes, Virtual function		
UNIT	TITLE	PERIODS
IV	OPERATOR OVERLOADING AND TYPE CONVERSIONS	8
Defining operator overloading, overloading unary and binary operators, Operator overloading using friend function, Rules for Overloading operators - Type conversion.		
UNIT	TITLE	PERIODS
V	FILES ,TEMPLATES AND EXCEPTION HANDLING	9
File pointer and file processing, Sequential and random file access -Template function, Template classes - Exception handling mechanism ,Standard Template Library – Case Study: online Reservation system, Student Management System		

TOTALPERIODS	45

TEXT BOOKS:	
1.	E.Balaguruswamy “Object Oriented Programming with C++”, 6th edition, Tata McGraw Hill Education, 2015
2.	Bjarne Stroustrup ,”The C++ Programming Language”, 4th Edition, Addison Wesley, 2015

REFERENCE BOOKS:	
1.	Stanley B., Lippman, JoseeLajoie and Barbara E. Moo, —C++ Primer, Pearson Education, New Delhi,2010.
2.	Herbert Schildt, —C++: The Complete Referencell, Tata McGraw Hill, New Delhi, 2009
3.	Yashwant Kanetkar, —Object Oriented Programming with C++, BPB Publications, New Delhi, 2004

WEBSITES:	
1.	http://nptel.ac.in/

JOURNALS:	
1.	Nil

EXTENSIVE READER:	
1.	Nil

COURSE DESIGNERS			
1.	Dr Ahamed Ali S	Assistant Professor	Computer Science and Engineering
2.	Dr Arockia Abins A	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	1
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC303T	FOUNDATIONS OF DATA SCIENCE (Common to CSE (AI & ML), CSE (CS), CSE)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To understand the data science fundamentals and process.
2.	To learn to describe the data for the data science process.
3.	To Utilize NumPy, Pandas for Efficient Data Handling
4.	To utilize the Python libraries for Data Wrangling.
5.	To be equipped with R and Tableau

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Define the data science process	K3
CO2:	Apply different types of data description for data science process	K3
CO3:	Explore knowledge on relationships between data	K3
CO4:	Use the Python Libraries for Data Wrangling	K4
CO5:	Apply visualization Libraries in R and Tableau to interpret and explore data	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	2	2	-	-	-	1	1	1	2
CO2	2	1	-	1	1	-	-	-	2	1	1	2
CO3	2	2	1	2	2	1	1	-	1	2	1	3
CO4	3	2	2	2	2	-	-	-	1	1	2	2
CO5	2	2	1	2	2	-	-	-	1	1	1	2

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	IMPORTANCE OF DATA SCIENCE, STATISTICS and PROBABILITY	9
Importance of Data Science: Need for Data Science - Data Science Process – components of Data Science – Statistics and Probability: Data Types – Variable Types – Statistics – Sampling Techniques and Probability – Information Gain and Entropy – Probability Theory – Probability Types – Probability Distribution Functions – Bayes' Theorem – Inferential Statistics		
UNIT	**TITLE**	**PERIODS**
II	DATA SCIENCE METHODOLOGY AND MACHINE LEARNING	9
Data Science Methodology : Analytics for Data Science – Examples of Data Analytics – Data Analytics Life Cycle - Regression Analysis : Linear Regression – Logistic Regression – Time Series Models- Machine Learning : Decision Trees – Naïve Bayes – Support Vector Machines – Nearest Neighbor Learning - Clustering		
UNIT	**TITLE**	**PERIODS**
III	NUMPY & PANDAS BASICS	9
NumPy : ndarray – Data Processing using Arrays – File input and Output with Arrays – Linear Algebra – Random Number Generation – Pandas : Pandas Data Structures – Essential Functionality – Summarizing and Computing Descriptive Statistics – Handling Missing Data – Hierarchical Indexing		
UNIT	**TITLE**	**PERIODS**
IV	DATA WRANGLING , PLOTTING & VIZUALIZATION	9
Data Wrangling : Combining and Merging Datasets – Reshaping and Pivoting – Data Transformation – String Manipulation – Plotting and Visualization : Plotting using Matplotlib – Plotting Function in Pandas – Plotting Maps		
UNIT	**TITLE**	**PERIODS**
V	DATA SCIENCE TOOLS: R AND TABLEAU	9
R : Reading and getting data into R – Ordered and Unordered Factors – Arrays and Matrices – Lists and Data Frames – Probability Distributions – Statistical Models in R – Manipulating Objects – Data Distribution – Data Visualization using Tableau : Dimensions and Measures, Descriptive Statics – Basic Charts – Dashboard Design & Principles – Special Chart Types		

TOTAL PERIODS:	45

TEXT BOOKS:	
1.	Sanjeev J. Wagh — Fundamentals of Data Science, CRC Press, First Edition, 2022.(Unit 1,2,5)
2.	Wes McKinney — Python for Data Analysis, O'Reilly Media, First Edition, 2012 (Unit 3,4)

REFERENCE BOOKS:	
1.	Hadley Wickham and Garrett Grolemund — R for Data Science, O'Reilly Media, First Edition, 2016
2.	Foster Provost and Tom Fawcett — Data Science for Business, O'Reilly Media, First Edition, 2013
3.	Jake VanderPlas — Python Data Science Handbook, O'Reilly Media, First Edition, 2016

WEBSITES:	
1.	https://www.datacamp.com/
2.	https://padhai.onefourthlabs.in/courses/data-science

JOURNALS:	
1.	Data Science and Engineering (https://link.springer.com/journal/41019)

EXTENSIVE READER:	
1.	The Data Science Handbook, Advice And Insights From 25 Amazing Data Scientists

COURSE DESIGNERS			
1.	Mr. S.Kingsley	Assistant Professor	Computer Science and Engineering
2.	Dr Arockia Abins A	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	1
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC311L	DATA STRUCTURES LABORATORY (Common to AI & DS, CSE (AI & ML), CSE (CS), CSE, IT)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		0	0	4	0	2

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	60	40	100	45	50

COURSE OBJECTIVES:	
1.	To understand and implement Linear data structures using C
2.	To implement Non Linear data structures using C
3.	To Implement of Graph and Traversal algorithms
4.	To implement Searching and Sorting algorithms
5.	To implement Hashing techniques

COURSE OUTCOMES:		Bloom's Level
Upon completion of this course, student will be able to:		
CO1:	Write functions to implement linear data structure operations using C	K3
CO2:	Solve problems using non-linear data structures	K3
CO3:	Implement Graph and Traversal algorithms	K3
CO4:	Develop searching and sorting algorithms	K3
CO5:	Develop programs using Hashing techniques	K3

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	1	1	-	3	3	-	2
CO2	3	3	3	3	2	2	2	-	2	2	-	2
CO3	3	3	3	3	2	2	2	-	2	2	-	2
CO4	3	3	3	3	2	2	2	-	2	2	-	2
CO5	3	3	3	2	2	1	1	-	3	-	-	2

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

LIST OF EXPERIMENTS	
1.	Array implementation of List ADT
2.	Array implementation of Stacks
3.	Array implementation of Queues
4.	Linked list implementation of List ADT

5.	Linked list implementation of Stacks
6.	Linked list implementation of Queues
7.	Application of Stacks and Queues
8.	Implementation of Binary Search Trees and Traversal
9.	Implementation of AVL Trees
10.	Implementation of Heaps using Priority Queues
11.	Implementation of Graph and Traversal algorithms
12.	Implementation of Sorting Algorithms : Bubble sort & Quick sort
13.	Implementation of Linear search and Binary search
14.	Implementation of Hashing - any one collision resolution techniques
TOTAL PERIODS	**60**

CONTENT BEYOND SYLLABUS	
1.	Implementation of Priority Queue
2.	Implementation of Red Black Tree

LIST OF EQUIPMENTS:	
1.	**HARDWARE**:
	• Standalone desktops.
2.	**SOFTWARE**
	• Linux Operating System with GCC / TURBO C in WINDOWS OS / TURBO C++ in WINDOWS OS

COURSE DESIGNERS			
1.	Mrs. K. HemaPriya	Assistant Professor	Computer Science and Engineering
2.	Dr.J.Deepa	Associate professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	1
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC312L	OBJECT ORIENTED PROGRAMMING LABORATORY	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**0**	**0**	**3**	**1**	**2**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	60	40	100	45	50

COURSE OBJECTIVES:	
1.	To use object oriented programming to Develop Generic programming skills
2.	Apply appropriate data structures and solve complex problems
3.	Design problem into classes and develop a full working code
4.	Develop programs using files, templates and handle exceptions
5.	Develop an application based on OOPs Concept

COURSE OUTCOMES:		
Upon completion of this course, student will be able to:		Bloom's level
CO1:	Develop programs using functions and constructors in C++.	K5
CO2:	Write programs using inheritance and polymorphism.	K3
CO3:	Implement the concepts of operator overloading in C++.	K4
CO4:	Develop programs using template for searching and sorting algorithms.	K6
CO5:	Design applications for file manipulation and exception handling.	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	3	2	-	-	-	2	-	2	3
CO2	3	3	3	2	2	-	-	-	2	-	2	3
CO3	3	3	3	2	2	-	-	-	2	-	2	3
CO4	3	3	3	2	2	-	-	-	2	-	2	3
CO5	3	3	3	3	1	2	2	1	2	2	2	3
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

LIST OF EXPERIMENTS	
1.	Design C++ classes with static members, methods with default arguments, friend functions. (For example, design matrix and vector classes with static allocation, and a friend function to do matrix-vector multiplication)
2.	Implement complex number class with necessary operator overloading's and type conversions such as integer to complex, double to complex, complex to double etc.
3.	Implement Matrix class with dynamic memory allocation and necessary methods. Give proper constructor,

	destructor, copy constructor, and overloading of assignment operator.
4.	Overload the new and delete operators to provide custom dynamic allocation of memory.
5.	Develop a template of linked-list class and its methods.
6.	Develop templates of standard sorting algorithms such as bubble sort, insertion sort, merge sort, and quick sort.
7.	Design stack and queue classes with necessary exception handling.
8.	Define Point class and an Arc class. Define a Graph class which represents graph as a collection of Point objects and Arc objects. Write a method to find a minimum cost spanning tree in a graph.
9.	Develop with suitable hierarchy, classes for Point, Shape, Rectangle, Square, Circle, Ellipse, Triangle, Polygon, etc. Design a simple test application to demonstrate dynamic polymorphism and RTTI.
10.	Write a C++ program that randomly generates complex numbers (use previously designed Complex class) and writes them two per line in a file along with an operator (+, -, *, or /). The numbers are written to file in the format (a + ib). Write another program to read one line at a time from this file, perform the corresponding operation on the two complex numbers read, and write the result to another file (one per line).
11.	MINI PROJECT: Create a —Railway reservation system / Airline reservation system with the following modules – Booking – Availability checking – Cancellation
	TOTAL PERIODS: 60

CONTENT BEYOND SYLLABUS	
1.	Develop an application by implementing Inheritance
2.	Develop an application using polymorphism

LIST OF EQUIPMENTS:	
1.	**HARDWARE**:
	• Standalone desktops.
2.	**SOFTWARE**
	• Linux Operating System with GCC / TURBO C in WINDOWS OS / TURBO C++ in WINDOWS OS

COURSE DESIGNERS			
1.	Dr Ahamed Ali S	Assistant Professor	Computer Science and Engineering
2.	Dr Arockia Abins A	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	1
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC313L	DATA SCIENCE LABORATORY (Common to CSE (AI & ML), CSE (CS), CSE)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**0**	**0**	**4**	**0**	**2**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	60	40	100	45	50

COURSE OBJECTIVES:	
1.	To understand the python libraries for data science
2.	To understand the basic Statistical and Probability measures for data science
3.	To learn descriptive analytics on the benchmark data sets
4.	To apply correlation and regression analytics on standard data sets
5.	To present and interpret data using visualization packages in Python

COURSE OUTCOMES:		Bloom's Level
Upon completion of this course, student will be able to:		
CO1:	Apply the python libraries for data science	K3
CO2:	Implement the basic Statistical and Probability measures for data science	K3
CO3:	Perform descriptive analytics on the benchmark data sets.	K3
CO4:	Perform correlation and regression analytics on standard data sets	K3
CO5:	Present and interpret data using visualization packages in Python	K3

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	1	1	-	3	3	-	2
CO2	3	3	3	3	2	2	2	-	2	2	-	2
CO3	3	3	3	3	2	2	2	-	2	2	-	2
CO4	3	3	3	3	2	2	2	-	2	2	-	2
CO5	3	3	3	2	2	1	1	-	3	-	-	2

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

LIST OF EXPERIMENTS	
1.	Download, install and explore the features of NumPy, SciPy, Jupyter, Statsmodels and Pandas packages
2.	Working with Numpy arrays
3.	Working with Pandas data frames

4.	Reading data from text files, Excel and the web and exploring various commands for doing descriptive analytics on the Iris data set
5.	Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following: a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis. b. Bivariate analysis: Linear and logistic regression modeling c. Multiple Regression analysis d. Also compare the results of the above analysis for the two data sets.
6.	Apply and explore various plotting functions on UCI data sets. a. Normal curves b. Density and contour plots c. Correlation and scatter plots d. Histograms e. Three dimensional plotting
7.	Visualizing Geographic Data with Basemap
	TOTAL PERIODS **60**

CONTENT BEYOND SYLLABUS	
1.	Using pandas for cleaning and preprocessing data, dealing with missing values, outliers, and data transformation.
2.	Implementing simple neural networks.

LIST OF EQUIPMENTS:	
1.	**HARDWARE**:
	• Standalone desktops.
2.	**SOFTWARE**
	• Python / Anaconda / Jupyter Notebook

COURSE DESIGNERS			
1.	Dr. S. Ahamed Ali	Assistant Professor	Computer Science and Engineering
2.	Mr. Kingsley. S	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	1
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

CURRICULUM AND SYLLABUS

(REGULATIONS 2023 V1.0)

FOR

B.E. COMPUTER SCIENCE AND ENGINEERING

CHOICE BASED CREDIT SYSTEM

(Applicable to the students admitted from the Academic Year 2023-24 onwards)



EASWARI ENGINEERING COLLEGE

(Autonomous Institution)
Bharathi Salai, Ramapuram, Chennai - 600 089

IV Sem

R2023 V1.0

SEMESTER IV									
S. No	Course Code	Course Title	Category	Hours / Week				CREDITS	Internal / External %
				L	T	P	R		
THEORY									
1	231MAB403T	Probability and Number Theory	BS	3	1	-	-	4	40 / 60
2	231CSC401T	Database Management Systems	PC	3	-	-	-	3	40 / 60
3	231CSC402T	Design and Analysis of Algorithms	PC	3	-	-	-	3	40 / 60
4	231CSC403T	Operating Systems	PC	3	-	-	-	3	40 / 60
5	231CSC404T	Java Programming	PC	3	-	-	-	3	40 / 60
LABORATORY									
6	231CSC411L	Database Management Systems Laboratory	PC	-	-	4	0	2	60 / 40
7	231CSC412L	Operating Systems Laboratory	PC	-	-	4	0	2	60 / 40
8	231CSC413L	Java Programming Laboratory	PC	-	-	3	1	2	60 / 40
9	231SDA002	Competitive Coding Skills	EEC	0	0	2	0	0	-
							TOTAL CREDITS	**22**	**-**

SYLLABUS OF
SEMESTER – IV
COURSES

231MAB403T	PROBABILITY AND NUMBER THEORY (Common to CSE, CSE-CS)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**1**	**0**	**0**	**4**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
NIL

COURSE OBJECTIVES:	
1.	To provide necessary basic concepts in probability and random processes for applications such as random signals, linear systems in communication engineering
2.	To understand the basic concepts of probability, one- and two-dimensional random variables and to introduce some standard distributions applicable to engineering which can describe real life phenomenon
3.	To understand the basic concepts in number theory
4.	To examine the key questions in the Theory of Numbers.
5.	To give an integrated approach to number theory and provide a firm basis for further reading and study in the subject.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Use the probability distribution to study discrete and continuous random variables.	K4
CO2:	Find the joint probability density function (PDF) of two new random variables by using the PDF of two given random variables and given transformation.	K5
CO3:	Apply division algorithm and Euclidean algorithm to find the GCD of any two positive integers.	K3
CO4:	Solve a system of linear congruences by using basic number theory results.	K5
CO5:	Solve linear congruences with the help of classical theorems on number theory.	K6

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	-	-	-	-	-	-	-	1
CO2	3	1	2	-	-	-	-	-	-	-	-	2
CO3	3	2	3	-	-	-	-	-	-	-	-	2
CO4	3	2	2	-	-	-	-	-	-	-	-	1
CO5	3	3	2	-	-	-	-	-	-	-	-	1
CO6	3	2	1	-	-	-	-	-	-	-	-	1

3 – High: 2 - Medium: 1 – Low: '-' - No correlation:

UNIT	TITLE	PERIODS
I	**PROBABILITY AND RANDOM VARIABLES**	**L - 8, T - 4**
Probability review – Baye's theorem, Discrete and continuous random variables – Moments – Moment generating functions – Binomial, Poisson, Geometric, Uniform, Exponential and Normal distributions.		
UNIT	**TITLE**	**PERIODS**
II	**TWO - DIMENSIONAL RANDOM VARIABLES**	**L - 8, T - 4**
Joint distributions – Marginal and conditional distributions – Covariance – Correlation and linear regression – Transformation of random variables.		
UNIT	**TITLE**	**PERIODS**
III	**DIVISIBILITY THEORY AND CANONICAL DECOMPOSITIONS**	**L - 8, T - 4**
Division algorithm – Base - b representations – Number patterns – Prime and composite numbers – GCD – Euclidean algorithm – Fundamental theorem of arithmetic – LCM.		
UNIT	**TITLE**	**PERIODS**
IV	**DIOPHANTINE EQUATIONS AND CONGRUENCES**	**L - 8, T - 4**
Linear Diophantine equations – Congruence's – Linear Congruence's - Applications: Divisibility tests - Modular exponentiation-Chinese remainder theorem – 2 x 2 linear systems.		
UNIT	**TITLE**	**PERIODS**
V	**CLASSICAL THEOREMS AND MULTIPLICATIVE FUNCTIONS**	**L - 8, T - 4**
Wilson's theorem - Fermat's little theorem – Euler's theorem – Euler's Phi functions – Tau and Sigma functions.		

TOTAL PERIODS:	**60**

COURSE OUTCOMES:	
Upon completion of this course, student will be able to:	
CO1:	Recall the concepts of probability to study discrete and continuous distribution.
CO2:	Compose the joint probability density function (PDF) of two new random variables by using the PDF of two given random variables and given transformation.
CO3:	Apply division algorithm and Euclidean algorithm to find the GCD of any two positive integers.
CO4:	Analyze linear congruence's to solve a system of linear congruence's.
CO5:	Revise classical theorems on number theory which help to solve linear congruence's.

TEXT BOOKS:	
1.	Koshy, T., —Elementary Number Theory with Applications‖, Elsevier Publications, New Delhi, 2002.
2.	Ibe. O.C., "Fundamentals of Applied Probability and Random Processes", Elsevier, 1st Indian Reprint, 2007.

WEBSITES	
1.	https://math.stackexchange.com
2.	https://link.springer.com
3.	https://sciencedirect.com

REFERENCE BOOKS:	
1.	Hwei Hsu, "Schaum's Outline of Theory and Problems of Probability, Random Variables and Random Processes", Tata McGraw Hill Edition, New Delhi, 2004.
2.	Niven, I., Zuckerman.H.S., and Montgomery, H.L., —An Introduction to Theory of Numbersǁ, John Wiley and Sons, Singapore, 2004.
3.	San Ling and Chaoping Xing, —Coding Theory – A first Courseǁ, Cambridge Publications, Cambridge, 2004.

COURSE DESIGNERS			
1.	Dr.S.Muthukumar	Professor	Department of Mathematics
2.	Dr.S.R.Ananthalakshmi	Associate Professor & HOD	Department of Mathematics
3.	Mrs.V.Sowmiya	Assistant Professor	Department of Mathematics

Recommended by Board of Studies	Date: 12.10.2023	Syllabus version	V.1
Approved by the Academic Council	Date: 24.01.2024	Meeting No.	6

231CSC401T	**DATABASE MANAGEMENT SYSTEMS** (Common to AI & DS, CSE, CSE (AI & ML), CSE (CS), IT)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To learn the fundamentals of data models, relational algebra and SQL.
2.	To represent a database system using ER diagrams and to learn normalization techniques.
3.	To understand the fundamental concepts of transaction, concurrency and recovery processing.
4.	To understand the internal storage structures using different file and indexing techniques which will help in physical DB design.
5.	To have an introductory knowledge about the Distributed databases, NOSQL and database security.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Develop proficiency in designing, implementing, and querying relational databases using SQL.	K3
CO2:	Design database using ER model and normalize the database.	K3
CO3:	Construct queries to handle transaction processing and maintain consistency of the database.	K3
CO4:	Analyze indexing strategies for File organization and Query Optimization.	K4
CO5:	Evaluate the distributed database and security challenges.	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	2	1	-	-	-	2	1	1	1
CO2	3	3	3	2	1	-	-	-	2	2	2	2
CO3	3	3	3	2	1	-	-	-	1	1	1	1
CO4	3	3	3	2	1	-	-	-	1	1	1	1
CO5	3	3	3	2	1	-	-	-	2	2	2	2
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

UNIT	TITLE	PERIODS
I	**RELATIONAL DATABASES**	10
Purpose of Database System – Views of data – Data Models – Database System Architecture – Introduction to relational databases – Relational Model – Keys – Relational Algebra – SQL fundamentals – Advanced SQL features – Embedded SQL– Dynamic SQL.		
UNIT	**TITLE**	**PERIODS**
II	**DATABASE DESIGN**	8
Entity-Relationship model – E-R Diagrams – Enhanced-ER Model – ER-to-Relational Mapping – Normalization: Functional Dependencies – Non-loss Decomposition – First, Second, Third Normal Forms, Dependency Preservation – Boyce/Codd Normal Form – Multi-valued Dependencies and Fourth Normal Form – Join Dependencies and Fifth Normal Form.		
UNIT	**TITLE**	**PERIODS**
III	**TRANSACTIONS**	9
Transaction Concepts – ACID Properties – Schedules – Serializability – Transaction support in SQL – Need for Concurrency – Concurrency control –Two Phase Locking- Timestamp – Deadlock Handling – Recovery Concepts – Recovery based on deferred and immediate update.		
UNIT	**TITLE**	**PERIODS**
IV	**IMPLEMENTATION TECHNIQUES**	9
RAID – File Organization – Organization of Records in Files – Data dictionary Storage – Indexing and Hashing –Ordered Indices – B+ tree Index Files – B tree Index Files – Static Hashing – Dynamic Hashing – Query Processing Overview – Query optimization using Heuristics - Cost Estimation.		
UNIT	**TITLE**	**PERIODS**
V	**ADVANCED TOPICS**	9
Distributed Databases: Architecture, Data Storage, Transaction Processing – NOSQL Databases- Data Mining and warehousing - Information retrieval - Database Security: Security issues – Access control based on privileges – Role Based access control.		
	TOTAL PERIODS:	45

TEXT BOOKS:	
1.	Abraham Silberschatz, Henry F. Korth, S. Sudharshan, “Database System Concepts”, Seventh Edition, McGraw Hill, 2020.
2.	RamezElmasri, Shamkant B. Navathe, “Fundamentals of Database Systems”, Seventh Edition, Pearson Education, 2017.

REFERENCE BOOKS:	
1.	C.J.Date, A.Kannan, S.Swamynathan, “An Introduction to Database Systems”, Eighth Edition, Pearson

	Education, 2006.
2.	Thomas M. Connolly, Carolyn E. Begg, "Database Systems – A Practical Approach to Design, Implementation, and Management", Sixth Edition, Global Edition, Pearson Education, 2015.
3.	Pramod J. Sadalage, Martin Fowler," NoSQL Distilled- A Brief Guide to the Emerging World of Polyglot Persistence", 2012

COURSE DESIGNERS			
1.	Dr.M.Mohana	Associate Professor	Information Technology
2.	Dr.S.Gnanapriya	Assistant Professor	Information Technology
3.	Dr.J.Deepa	Associate Professor	Computer Science and Engineering
4.	Dr.A.Jeba Sheela	Associate Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	V1.0
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC402T	**DESIGN AND ANALYSIS OF ALGORITHMS** (Common to AI & DS, CSE, CSE (AI & ML), CSE (CS))	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
NIL

COURSE OBJECTIVES:	
1.	To demonstrate performance of algorithms with respect to time and space complexity
2.	To perform various computational complexities using Brute Force and Divide-And-Conquer techniques.
3.	To critically analyze the efficiency of various algorithms using dynamic programming and greedy method
4.	To critically analyze the efficiency of alternative algorithmic solutions for the same problem
5.	To familiarize the concept of deterministic and non deterministic algorithm

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Analyze the algorithms by time and space complexity	K3
CO2:	Solve the problems using Brute force and Divide-and-Conquer method	K3
CO3:	Solve the problems using Dynamic Programming And Greedy technique	K3
CO4:	Build the solution for the problem using Iterative algorithmic design techniques	K4
CO5:	Design algorithms for the real-world problems	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	3	2	-	-	-	2	-	2	3
CO2	3	3	3	2	2	-	-	-	2	-	2	3
CO3	3	3	3	2	2	-	-	-	2	-	2	3
CO4	3	3	3	2	2	-	-	-	2	-	2	3
CO5	3	3	3	3	1	2	2	1	2	2	2	3

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	**INTRODUCTION**	9
Notion of an Algorithm – Fundamentals of Algorithmic Problem Solving – Important Problem Types – Fundamentals of the Analysis of Algorithmic Efficiency –Asymptotic Notations and their properties. Analysis Framework - Mathematical analysis for Recursive and Non-recursive algorithms – Visualization		
UNIT	**TITLE**	**PERIODS**
II	**BRUTE FORCE AND DIVIDE-AND-CONQUER**	9
Brute Force – Computing a^n – String Matching - Closest-Pair and Convex-Hull Problems - Exhaustive Search - Travelling Salesman Problem - Knapsack Problem - Assignment problem. Divide and Conquer Methodology – Binary Search – Merge sort – Quick sort – Heap Sort - Multiplication of Large Integers.		
UNIT	**TITLE**	**PERIODS**
III	**DYNAMIC PROGRAMMING AND GREEDY TECHNIQUE**	9
Dynamic programming – Principle of optimality - Coin changing problem, Computing a Binomial Coefficient – Floyd's algorithm – Multi stage graph - Optimal Binary Search Trees – Knapsack Problem and Memory functions. Greedy Technique – Container loading problem - Prim's algorithm and Kruskal's Algorithm – 0/1 Knapsack problem, Huffman Trees.		
UNIT	**TITLE**	**PERIODS**
IV	**ITERATIVE IMPROVEMENT**	9
The Simplex Method - The Maximum-Flow Problem – Maximum Matching in Bipartite Graphs, Stable marriage Problem.		
UNIT	**TITLE**	**PERIODS**
V	**COPING WITH THE LIMITATIONS OF ALGORITHM POWER**	9
Lower - Bound Arguments - P, NP NP-Complete and NP Hard Problems. Backtracking – n-Queen's problem - Hamiltonian Circuit Problem – Subset Sum Problem Branch and Bound – Assignment problem – Knapsack Problem – Travelling Salesman Problem - Approximation Algorithms for NP- Hard Problems – Travelling Salesman problem – Knapsack problem.		

TOTAL PERIODS:	**45**

COURSE OUTCOMES:	
Upon completion of this course, student will be able to:	
CO1:	Analyze the algorithms by time and space complexity.
CO2:	Solve the problems using Brute force and Divide-and-Conquer method.
CO3:	Solve the problems using Dynamic Programming and Greedy technique.
CO4:	Build the solution for the problem using Iterative algorithmic design techniques.
CO5:	Design algorithms for the real-world problems.

TEXT BOOKS:	
1.	Anany Levitin," Introduction to the Design and Analysis of Algorithms", Third Edition, Pearson Education,2012.
2.	Ellis Horowitz, SartajSahni and SanguthevarRajasekaran, Computer Algoritms / C++,Second Edition, University Press,2007.

REFERENCE BOOKS:	
1.	Alfred V. Aho, John E.Hopcroft and Jeffrey D.Ullman, ”Data Structures and Algorithms”, Pearson Education, Reprint 2006.
2.	Harsh Bhasin, ”Algorithms Design and Analysis”,Oxford university press,2016.
3.	Sridhar S ,”Design and Analysis of Algorithms”,Oxford university press,2014.
4.	Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, ”Introduction to Algorithms”, Third Edition,PHI Learning Private Limited,2012

WEBSITES:	
1.	http://nptel.ac.in/

COURSE DESIGNERS			
1.	B.S.Liya	Assistant Professor	Computer Science and Engineering
2.	S.Maheswari	Assistant Professor	Computer Science and Engineering
3.	P.Indumathy	Assistant Professor	Computer Science and Engineering
4.	K.Shankar	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	V1.0
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC403T	**OPERATING SYSTEMS** (Common to AI & DS, CSE, CSE (AI & ML), CSE (CS))	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
Nil

COURSE OBJECTIVES:	
1.	To understand the basics and functions of operating systems.
2.	To understand processes and threads.
3.	To analyze scheduling algorithms and process synchronization.
4.	To understand the concept of deadlocks.
5.	To analyze various memory management schemes.
6	To be familiar with I/O management and file systems.
7.	To be familiar with the basics of virtual machines and Mobile OS like iOS and Android.

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Analyze various scheduling algorithms and process synchronization.	K3
CO2:	Explain deadlock prevention and avoidance algorithms.	K3
CO3:	Compare and contrast various memory management schemes.	K3
CO4:	Explain the functionality of file systems, I/O systems, and Virtualization.	K4
CO5:	Compare iOS and Android Operating Systems.	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	2	1	-	3	-	-	-	3	3	3	3
CO2	3	3	3	3	3	-	-	-	3	3	3	3
CO3	3	3	3	3	3	-	-	-	3	3	3	3
CO4	3	3	3	2	1	-	-	-	3	1	2	3
CO	3	3	3	3	-	2	2	1	2	-	1	3

5												
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

UNIT	TITLE	PERIODS
I	**INTRODUCTION**	**7**
Computer System - Elements and organization; Operating System Overview - Objectives andFunctions - Evolution of Operating System; Operating System Structures – Operating System Services - User Operating System Interface - System Calls – System Programs - Design and Implementation - Structuring methods.		
UNIT	**TITLE**	**PERIODS**
II	**PROCESS MANAGEMENT**	**11**
Processes - Process Concept - Process Scheduling - Operations on Processes - Inter-process Communication; CPU Scheduling - Scheduling criteria - Scheduling algorithms: Threads -Multithread Models – Threading issues; Process Synchronization - The Critical-Section problem -Synchronization hardware – Semaphores – Mutex - Classical problems of synchronization -Monitors; Deadlock - Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock.		
UNIT	**TITLE**	**PERIODS**
III	**MEMORY MANAGEMENT**	**10**
Main Memory - Swapping - Contiguous Memory Allocation – Paging - Structure of the Page Table -Segmentation, Segmentation with paging; Virtual Memory - Demand Paging – Copy on Write – Page Replacement - Allocation of Frames –Thrashing.		
UNIT	**TITLE**	**PERIODS**
IV	**STORAGE MANAGEMENT**	**10**
Mass Storage system – Disk Structure - Disk Scheduling and Management; File-System Interface -File concept - Access methods - Directory Structure - Directory organization - File system mounting- File Sharing and Protection; File System Implementation - File System Structure – Directoryimplementation - Allocation Methods - Free Space Management; I/O Systems – I/O Hardware, Application I/O interface, Kernel I/O subsystem.		
UNIT	**TITLE**	**PERIODS**
V	**VIRTUAL MACHINES AND MOBILE OS**	**7**
Virtual Machines – History, Benefits and Features, Building Blocks, Types of Virtual Machines and their Implementations, Virtualization and Operating-System Components; Mobile OS - iOS andAndroid		

TOTAL PERIODS:	**45**

TEXT BOOKS:	
1.	Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Concepts" ll, 10th Edition, John Wiley and Sons Inc., 2018.
2.	Andrew S Tanenbaum, "Modern Operating Systems", Pearson, 5th Edition, 2022 New Delhi.
REFERENCE BOOKS:	
1.	RamazElmasri, A. Gil Carrick, David Levine, " Operating Systems – A Spiral Approach",Tata McGraw Hill Edition, 2010.
2.	William Stallings, "Operating Systems: Internals and Design Principles", 7th Edition,

	Prentice Hall, 2018.
3.	AchyutS.Godbole, AtulKahate, “Operating Systems”, McGraw Hill Education, 2016.

WEBSITES:	
1.	https://www.studytonight.com/operating-system/
2.	https://prepinsta.com/operating-systems/

COURSE DESIGNERS			
1.	Dr.P. K. Devan	Assistant Professor	Computer Science and Engineering
2.	Mr. K. Shankar	Assistant Professor	Computer Science and Engineering
3.	Ms. B. S. Liya	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	V1.0
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231CSC404T	**JAVA PROGRAMMING** (B.E Computer Science Engineering)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**3**	**0**	**0**	**0**	**3**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	40	60	100	45	50

PREREQUISITES:
Programming in C

COURSE OBJECTIVES:	
1.	To understand Object Oriented Programming concepts and basic characteristics of Java
2.	To know the principles of packages, inheritance and interfaces
3.	To define exceptions and use I/O streams
4.	To develop a java application with threads and generics classes
5.	To design and build simple Graphical User Interfaces

COURSE OUTCOMES (COs):		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Develop Java programs using OOP principles	K5
CO2:	Develop Java programs with the concept's inheritance and Interfaces	K5
CO3:	Build Java applications using exceptions and I/O streams	K6
CO4:	Develop Java applications with threads and generics classes	K3
CO5:	Develop interactive Java programs using JavaFX	K5

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	3	2	-	-	-	2	-	2	3
CO2	3	3	3	2	2	-	-	-	2	-	2	3
CO3	3	3	3	2	2	-	-	-	2	-	2	3
CO4	3	3	3	2	2	-	-	-	2	-	2	3
CO5	3	3	3	3	1	2	2	1	2	2	2	3

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

UNIT	TITLE	PERIODS
I	**INTRODUCTION TO OOP AND JAVA FUNDAMENTALS**	9
Object Oriented Programming - Abstraction – objects and classes -Encapsulation- Inheritance - Polymorphism- Characteristics of Java –The Java Environment - Java Source File - Compilation. FundamentalProgramming Structures in Java – Defining classes in Java–constructors, methods - Access specifiers - static members -Comments, Data Types, Variables, Operators, Control Flow, Arrays, Packages - JavaDoc comments.		
UNIT	**TITLE**	**PERIODS**
II	**INHERITANCE AND INTERFACES**	9
Inheritance – Super classes- sub classes –Protected members –constructors in sub classes- the Object class – abstract classes and methods- final methods and classes – Interfaces – defining an interface, implementing interface, extending interfaces – Differences between classes and interfaces - Object cloning - Inner classes- Strings-StringBuffer.		
UNIT	**TITLE**	**PERIODS**
III	**EXCEPTION HANDLING AND I/O**	9
Exception Handling: Definition of an Exception-Exception Classes- Common Exceptions; Exception Handling Techniques-Throwing and catching exceptions-Creating own exceptions- Streams-Byte Stream and Character stream-Files-Reading and writing files.		
UNIT	**TITLE**	**PERIODS**
IV	**MULTITHREADING AND GENERIC PROGRAMMING**	9
Differences between multi-threading and multitasking -Thread lifecycle - Creating threads - Synchronizing threads - Inter-thread communication - Daemon threads - Thread groups. Generic Programming – Generic classes – Generic methods – Bounded Types– Restrictions and Limitations.		
UNIT	**TITLE**	**PERIODS**
V	**JAVAFX EVENT HANDLING, CONTROLS AND COMPONENTS**	9
JAVAFX Events and Controls: Event Basics – Handling Key and Mouse Events. Controls: Checkbox, ToggleButton – RadioButtons – ListView – ComboBox – ChoiceBox – Text Controls –ScrollPane. Layouts		

TOTAL PERIODS:	**45**

TEXT BOOKS:	
1.	Cay S. Horstmann, Gary cornell, “Core Java Volume –IFundamentals”, 9th Edition, Prentice Hall, 2013.
2.	Herbert Schildt, “Java The complete reference”, 8th Edition,McGraw Hill Education, 2011.

REFERENCE BOOKS:	
1.	Danny Poo, Derek Kiong, Swarnalatha Ashok, “Object-Oriented Programming and Java”, 2nd Edition, SpringerPublication, 2008.
2.	Paul Deitel, Harvey Deitel, “Java SE 8 for programmers”, 3rdEdition, Pearson, 2015.
3.	Steven Holzner, “Java 2 Black book”, Dreamtech press, 2011
4.	Timothy Budd, “Understanding Object-oriented programmingwith Java”, Updated Edition, Pearson Education, 2000.

WEBSITES:	
1.	http://nptel.ac.in/
2.	https://www.w3schools.com/java/

COURSE DESIGNERS			
1.	Dr Ahamed Ali	Assistant Professor	Computer Science and Engineering
2.	Dr ArockiaAbins	Assistant Professor	Computer Science and Engineering
3	MsIndhumathy	Assistant Professor	Computer Science and Engineering
4	Mr Kingsley	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	V1.0
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231CSC411L	**DATABASE MANAGEMENT SYSTEMS LABORATORY** (Common to AI & DS, CSE, CSE (AI & ML), CSE (CS))	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**0**	**0**	**3**	**1**	**2**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	60	40	100	45	50

COURSE OBJECTIVES:	
1.	To learn and implement important commands in SQL.
2.	To learn the usage of nested and joint queries.
3.	To understand functions, procedures and procedural extensions of databases.
4.	To understand design and implementation of typical database applications.
5.	To be familiar with the use of a front-end tool for GUI based application development.

COURSE OUTCOMES:		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Create databases with different types of key constraints.	K3
CO2:	Construct simple and complex SQL queries using DML and DCL commands.	K3
CO3:	Use advanced features such as stored procedures and triggers and incorporate in GUI based application development.	K3
CO4:	Create and manipulate data using NOSQL database.	K6
CO5:	Create database applications using Front End Tools with database connectivity.	K6

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	2	2	-	-	1	3	1	3	2
CO2	3	3	3	2	2	-	-	1	1	2	3	2
CO3	3	3	3	2	2	-	-	1	1	1	3	2
CO4	3	3	3	2	2	-	-	1	1	1	3	2
CO5	3	3	3	2	2	-	-	1	1	1	3	2
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

LIST OF EXPERIMENTS	
1.	Create a database table, add constraints (primary key, unique, check, Not null), insert rows, update and delete rows using SQL DDL and DML commands.
2.	Create a set of tables, add foreign key constraints and incorporate referential integrity.
3.	Query the database tables using different 'where' clause conditions and also implement aggregate functions.
4.	Query the database tables and explore sub queries and simple join operations.
5.	Query the database tables and explore natural, equi and outer joins.
6.	Write user defined functions and stored procedures in SQL.
7.	Execute complex transactions and realize DCL and TCL commands.
8.	Write SQL Triggers for insert, delete, and update operations in a database table.
9.	Create View and index for database tables with a large number of records.
10.	Create Document, column and graph-based data using NOSQL database tools
11.	Develop a simple GUI based database application and incorporate all the above-mentioned features.
12.	Case Study using any of the real-life database applications from the following list a) Inventory Management for a EMart Grocery Shop b) Society Financial Management c) Cop Friendly App – Eseva d) Property Management – eMall e) Star Small and Medium Banking and Finance ● Build Entity Model diagram. The diagram should align with the business and functional goals stated in the application. ● Apply Normalization rules in designing the tables in scope. ● Prepared applicable views, triggers (for auditing purposes), and functions for enabling enterprise grade features. ● Build PL SQL / Stored Procedures for Complex Functionalities, ex: EOD Batch Processing for calculating the EMI for Gold Loan for each eligible Customer. ● Ability to showcase ACID Properties with sample queries with appropriate settings
TOTAL PERIODS:	**60**

CONTENT BEYOND SYLLABUS	
1.	Create an XML database and validate it using XML schema.
2.	SQL injection.

LIST OF EQUIPMENTS:	
1.	**HARDWARE**:
	Standalone desktops.
2.	**SOFTWARE**
	Oracle, MySQL and Eclipse / NetBeans IDE or Equivalent

COURSE DESIGNERS			
1.	Dr.G.S.Anandha Mala	Professor	Computer Science and Engineering
2.	A.JebaSheela	Assistant Professor	Computer Science and Engineering
3.	A.Bhagyalakshmi	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	V1.0
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC412L	**OPERATING SYSTEMS LABORATORY** (Common to CSE, CSE (CS))	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**0**	**0**	**3**	**0**	**2**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	60	40	100	45	50

COURSE OBJECTIVES:	
1.	To write programs in Linux environment using system calls.
2.	To implement Process Creation and Inter Process Communication.
3.	To implement page replacement algorithms
4.	To implement File Organization and File Allocation Strategies
5.	To implement Disk Scheduling Algorithms.

COURSE OUTCOMES:		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Illustrate the use of system calls in operating systems	Understand
CO2:	Evaluate the performance of different types of CPU scheduling algorithms.	Apply
CO3:	Ability to design and solve synchronization problems.	Apply
CO4:	Ability to simulate and implement deadlock management	Apply
CO5:	Ability to simulate and implement file management and memory management.	Apply
CO6:	Implement Modules for Disk Scheduling	Apply

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	3	3	3	3	2	1	1	1	1	1	3
CO2	3	3	2	3	2	2	2	1	1	1	1	3
CO3	3	3	3	3	3	1	1	1	1	1	1	3
CO4	3	3	2	3	2	3	3	2	1	1	2	3
CO5	3	2	2	2	2	3	3	2	1	1	2	3
CO6	3	3	3	3	3	2	1	1	1	1	1	3

3 – High : 2 - Medium : 1 – Low : '-' - No correlation

LIST OF EXPERIMENTS	
1.	Basics of UNIX commands and Implementation of Shell Programming
2.	Write C programs using the following system calls of UNIX operating system fork, exec, getpid, exit, wait, close, stat, opendir, readdir
3.	Implementation of CPU Scheduling. (i) FCFS ii) SJF iii) Priority iv) Round Robin
4.	Implementation ofProducer-Consumer Problem using Semaphores and Reader Writer Problem
5.	Write C program to implement Threading & Synchronization Applications
6.	Develop a program to simulate the concept of Dining-Philosophers problem.
7.	Develop a program to simulate the concept of Deadlock avoidance and detection.
8.	Simulate the following contiguous memory allocation techniques a) Worst-fit b) Best-fit c) First-fit
9.	Develop a program to simulate paging technique of memory management
10.	Simulate the following page replacement algorithms: a) FIFO b) LRU c) Optimal d)LFU
11.	Simulate the following file allocation strategies: a) Sequential b) Indexed c) Linked
12.	Develop a program to simulate the following disk scheduling algorithms: a) FCFS b) SCAN c) C-SCAN
TOTAL PERIODS:	**60**

CONTENT BEYOND SYLLABUS	
1.	Write a C program to implement compaction technique.
2.	Write a C program to simulate LRU-approximation page replacement algorithm? a. Additional-Reference bits algorithm b. Second-chance algorithm

LIST OF EQUIPMENTS:	
1.	**HARDWARE**:
	Standalone desktops.
2.	**SOFTWARE**
	Unix/Linux Turbo C

COURSE DESIGNERS			
1.	Dr.P. K. Devan	Assistant Professor	Computer Science and Engineering
2.	Mr. K. Shankar	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	V1.0
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7

231CSC413L	**Java Programming Laboratory** (B.E Computer Science Engineering)	Periods per week				Credits
		L	T	P	R	
Regulation - R23 V1.0		**0**	**0**	**3**	**0**	**2**

SCHEME OF EXAMINATION					
Duration of End Semester Examination in Hours	Marks			Minimum marks for Pass	
	Continuous assessment Examination	End Semester Examination	Maximum marks	End Semester Examination	Total
3	60	40	100	45	50

COURSE OBJECTIVES:	
1.	To build software development skills using java programming for real-world applications.
2.	To understand and apply the concepts of classes, packages, interfaces, array list, exception handling and file processing.
3.	To develop applications using event handling.
4.	To understand the concept of multithreading.

COURSE OUTCOMES:		
Upon completion of this course, student will be able to:		Bloom's Level
CO1:	Develop java programs using object-oriented programming concepts	K4
CO2:	Develop simple applications using packages ,abstract classes and using exception handling principles	K4
CO3:	Develop applications using generic programming	K4
CO4:	Implement multithreaded applications	K3
CO5:	Create GUIs and event driven programming applications for real world problems	K6

MAPPING OF COURSE OUTCOMES (CO) WITH PROGRAMME OUTCOME (PO)												
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	2	1	2	1	-	-	-	-	1	2	2	2
CO2	2	1	3	1	-	-	-	-	2	3	3	2
CO3	2	2	1	2	1	-	-	-	1	2	1	3
CO4	2	2	1	3	-	-	-	-	3	1	1	1
CO5	1	3	3	1	3	-	-	-	1	1	1	1
3 – High : 2 - Medium : 1 – Low : '-' - No correlation												

LIST OF EXPERIMENTS	
1.	Develop a Java application to generate Electricity bill. Create a class with the following members: ✓ Consumer no., consumer name, previous month reading, current month reading, type of EB connection (i.e. domestic or commercial). Compute the bill amount using the following tariff. ✓ If the type of the EB connection is domestic, calculate the amount to be paid as follows: ✓ First100units - Rs. 1 per unit ✓ 101-200units - Rs. 2.50 per unit ✓ 201 -500 units - Rs. 4 per unit ✓ > 501units - Rs. 6 per unit ✓ If the type of the EB connection is commercial, calculate the amount to be paid as follows: ✓ First 100 units - Rs. 2 per unit ✓ 101-200 units - Rs. 4.50 per unit ✓ 201 -500 units - Rs. 6 per unit ✓ > 501 units - Rs. 7 per unit
2.	Write a java program to implement the concept of packages.
3.	Develop a java application with an Employee class with Emp_name, Emp_id, Address, Mail_id, Mobile_no as members. Inherit the classes, Programmer, Assistant Professor, Associate Professor and Professor from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club funds. Generate pay slips for the employees with their gross and net salary.
4.	Write a Java Program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method printArea( ) that prints the area of the given shape.
5.	Solve the above problem using an interface.
6.	Implement exception handling and creation of user-defined exceptions
7.	Write a java program that implements a multi-threaded application that has three threads. First thread generates a random integer every 1 second and if the value is even, the second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of the cube of the number.
8.	Develop a Library Management System using generic classes in Java. The system should manage different types of entities such as books, members, and transactions.
9.	Write a Java program to perform write and read operations in file handling
10.	Develop application using JavaFX controls, layouts and menus
	TOTAL PERIODS: 60

CONTENT BEYOND SYLLABUS	
1.	Create custom annotations for metadata, and process them using Java's **Annotation Processing API**.
2.	Create an application for basic image processing (grayscale conversion, resizing, filters).

LIST OF EQUIPMENTS:	
1.	**HARDWARE**:
	• Standalone desktops.

2.	SOFTWARE

- Windows / Linux OS
- NetBeans, java

COURSE DESIGNERS			
1.	Dr S Ahamed Ali	Assistant Professor	Computer Science and Engineering
2.	Dr ArockiaAbins	Assistant Professor	Computer Science and Engineering

Recommended by Board of Studies	Date: 10.05.2024	Syllabus version	V1.0
Approved by the Academic Council	Date: 22.07.2024	Meeting No.	7